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Well-quasi-ordered classes of bounded clique-width

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Abstract

We study classes of graphs with bounded clique-width that are well-quasi-ordered by the induced subgraph relation, in the presence of labels on the vertices. We prove that, given a finite presentation of a class of graphs, one can decide whether the class is labelled-well-quasi-ordered. This solves an open problem raised by Daligault, Rao and Thomassé in 2010, and answers positively to two conjectures of Pouzet in the restricted case of bounded clique-width classes. Namely, we prove that being labelled-well-quasi-ordered by a set of size 2 or by a well-quasi-ordered infinite set are equivalent conditions, and that in such cases, one can freely assume that the graphs are equipped with a total ordering on their vertices. Finally, we provide a structural characterization of those classes as those that are of bounded clique-width and do not existentially transduce the class of all finite paths.

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This document uses knowledge: a notion points to its *definition*.

1 Introduction

The theory of *well-quasi-orderings* (WQOs) provides a powerful combinatorial setting that has found applications in various areas of mathematics and computer science. In graph theory, the celebrated result of Robertson and Seymour [?] states that the class of all finite graphs is well-quasi-ordered under the minor relation, a profound result with deep algorithmic consequences. WQOs are also at the heart of *well-structured transition systems*, infinite-state transition systems over which verification algorithms can be designed [?, ?]. One appealing feature of WQOs is that they are closed under various operations, such as the sum and the product of WQOs. As an example, the fact that X^* (finite words having elements of X as letters) is a WQO whenever X is a WQO, also known as Higman's lemma [?], has been used in the verification of so-called *lossy channel systems* [?].

Undirected finite graphs are naturally equipped with the *induced subgraph relation*, where G is an induced subgraph of H if G can be obtained from H by deleting vertices. Unlike the graph minor relation, the induced subgraph relation is not a well-quasi-ordering on the class of all finite graphs, as witnessed by the infinite family of cycles of increasing size, which form an infinite sequence of pairwise incomparable graphs. However, some classes of finite graphs are well-quasi-ordered under the induced subgraph relation: for instance, the class of all finite paths, or the class of all finite cliques. Distinguishing which classes of (finite) graphs are well-quasi-ordered under the induced subgraph relation is a long-standing open

Class	WQO	2-WQO	∀-WQO
Cycles	×	×	×
Paths	✓	×	×
Cliques	✓	✓	✓

Figure 1 Well-quasi-ordering properties of classes in **??**. No example 2-WQO but not ∀-WQO is known.

problem in graph theory.

In general, there is little hope of characterising all such classes, since any countable order with finite infixes can be embedded into the induced subgraph relation on finite graphs (this is a consequence of [**?**, Lemma 5.1]). If one considers *labelled* well-quasi-orderings on classes of finite graphs, then there are high hopes of being able to provide a full characterisation of (labelled) well-quasi-ordered classes. Given a class $\mathcal{C}$ of finite graphs, and a well-quasi-ordering $(X, \leq)$, one can consider the class $\mathsf{Label}_X(\mathcal{C})$ of graphs in $\mathcal{C}$ whose vertices are labelled by elements of X, that is, equipped with a function $\ell\colon V(G) \to X$. The induced subgraph relation is then extended to labelled graphs by requiring that the labels are preserved by the embedding, i.e. that the label of a vertex in the smaller graph is less than or equal to the label of its image in the larger graph. A class $\mathcal{C}$ of finite graphs is said to be *$(X, \leq)$-well-quasi-ordered* if the class $\mathsf{Label}_X(\mathcal{C})$ is well-quasi-ordered under the labelled induced subgraph relation. We will say that it is *k-well-quasi-ordered* if it is $(X, \leq)$-well-quasi-ordered where X is a k-element antichain, and that it is *∀-well-quasi-ordered* if it is $(X, \leq)$-well-quasi-ordered for every well-quasi-ordering $(X, \leq)$.

▶ **Example 1.** The class Cycles of all finite cycles is not well-quasi-ordered, since the infinite sequence of cycles forms an infinite antichain. The class Paths of all finite paths is well-quasi-ordered since $(\mathsf{Paths}, \subseteq_i)$ is isomorphic to $(\mathbb{N}, \leq)$. However, it is not 2-well-quasi-ordered, since the infinite sequence of paths with coloured endpoints forms an infinite antichain. The class Cliques of all finite cliques is ∀-well-quasi-ordered, since any X-labelled clique can be identified with a finite multiset of labels from X, and the set of finite multisets over a well-quasi-order is itself well-quasi-ordered by standard results on well-quasi-orders [**?**].

The study of labelled structures is a recurring theme in the theory of well-quasi-orderings: classical results such as Higman's lemma [**?**] or Kruskal's tree theorem [**?**] are actually stating that some classes of finite relational structures (respectively finite linear orders and finite meet trees) are well-quasi-ordered under the labelled embedding relation, for every well-quasi-ordering of labels. It was conjectured by Pouzet [**?**] that being ∀-well-quasi-ordered reduces to being 2-well-quasi-ordered for hereditary classes of structures (see [**?**, Problems 9, (1)] for a modern formulation). Another interesting conjecture of Pouzet [**?**, Problems 12] states that when a class $\mathcal{C}$ is ∀-well-quasi-ordered, one can add a total order on the vertices of its structures such that the new class is still ∀-well-quasi-ordered. This conjecture gained interest in the recent development of algebraic methods for polynomials over relational structures [**?**, Section 3, Theorem 11 and 12], where "monomials" are defined as structures labelled by $(\mathbb{N}, \leq)$. We refer to this property as being *orderably ∀-well-quasi-ordered*.

Even for classes of finite undirected graphs, these conjectures remain open. Several works dating back to the 90s have identified structural properties of hereditary classes of finite graphs that guarantee that they are ∀-well-quasi-ordered, such as being of bounded tree-depth [**?**]. In 2010, Daligault, Rao and Thomassé [**?**] initiated the study in the reverse direction, by considering a syntax for graph classes (using NLC graph expressions), and identifying which ones among them are 2-well-quasi-ordered. In this context, they verified Pouzet's conjectures, but were unable to generalise their results to arbitrary graph subclasses [?, Conjecture 4]. All the classes considered in [?] have what is called *bounded clique-width*,[1] and it is believed that it is enough to understand hereditary classes of bounded clique-width to solve these conjectures in full generality, as stated in the following conjecture.

▶ **Conjecture 2** ([?, Conjecture 5]). *Let $\mathcal{C}$ be a hereditary class of finite graphs that is 2-well-quasi-ordered. Then, $\mathcal{C}$ is of bounded clique-width.*

In parallel with Pouzet's conjectures, another line of research has focused on finding structural obstructions to being $\forall$-well-quasi-ordered. The natural candidate for such an obstruction is the class of all finite paths (see **??**). For the particular classes studied in [?], it was shown that every class that is not 2-well-quasi-ordered contains all finite paths [?, Corollary 2]. There are examples of hereditary classes that are not 2-well-quasi-ordered and do not contain all finite paths, but in every known such example, one can "extract" finite paths using very simple logical formulae. This led to the following conjecture, where existentially transduces is a notion from model theory (see **??** for a formal definition).

▶ **Conjecture 3** ([?, Conjecture 26]). *Let $\mathcal{C}$ be a hereditary class of finite graphs. If $\mathcal{C}$ is not 2-well-quasi-ordered, then it existentially transduces the class of all finite paths.*

In 2024, Lopez [?] studied labelled well-quasi-orderings on classes of graphs of bounded *linear* clique-width, leveraging combinatorial tools from automata theory (Simon's factorisation forests [?]). This yielded a weakening of Pouzet's conjectures [?, Corollary 2], namely, that a hereditary class of bounded linear clique-width is $\forall$-well-quasi-ordered if and only if it is k-well-quasi-ordered for every $k \in \mathbb{N}$. It is apparent from the proof that the techniques also yield a natural ordering on the vertices whenever the class is $\forall$-well-quasi-ordered, although this is not formally stated in the paper. It was conjectured [?, Section 5] that these techniques could be refined to work on classes of bounded clique-width (not necessarily linear), and actually obtain the correspondence between 2-well-quasi-ordering and $\forall$-well-quasi-ordering.

Contributions

Our main contribution is **??**, which answers positively to the conjectures of [?]. To that end, we first consider a generalisation of the framework of [?] from NLC graph expressions to MSO-interpretations from trees (these will be formally defined in **??**). This also generalises the setting of [?], which studied MSO-interpretations from *words*. Our main combinatorial result is the following.

▶ **Theorem 4.** *Let $\mathcal{C}$ be the image of a class of finite trees under an* MSO *interpretation $\mathcal{I}$. Then, the following are equivalent and decidable given $\mathcal{I}$:*

1. *$\mathcal{C}$ is 2-well-quasi-ordered,*
2. *$\mathcal{C}$ is $\forall$-well-quasi-ordered,*
3. *$\mathcal{C}$ is orderably $\forall$-well-quasi-ordered,*

From **??**, we derive the following corollary that solves Pouzet's conjectures for hereditary classes of bounded clique-width. This trades the decidability of **??** to handle more classes.[2]

[1] This will be defined formally in **??**

[2] There are uncountably many hereditary classes of bounded clique-width, but only countably many images of trees under MSO interpretations.

▶ **Theorem 5.** *The conclusion of* **??** *holds when* $\mathcal{C}$ *is a hereditary class of graphs having bounded clique-width.*

Because our proofs are constructive and combinatorial, we can derive from them obstructions to being ∀-well-quasi-ordered, which we list in **??**. In our statements, we use infinite antichains with a periodic structure that were called *periodic antichains* in [**?**, Section 7]. These antichains appear in [**?**, Conjecture 2], that is concerned with classes of graphs defined by finitely many excluded induced subgraphs, and relates the presence of these antichains to the fact that they are well-quasi-ordered. We show that in our setting, containing such periodic antichains is equivalent to being not 2-well-quasi-ordered. The precise definitions of regular antichains and periodic antichains will be given in **??**.

▶ **Theorem 6.** *Let* $\mathcal{C}$ *be a hereditary class of finite graphs of bounded clique-width. Then, the following are equivalent:*

1. $\mathcal{C}$ *is not* 2*-well-quasi-ordered,*
2. *There exists a class* $\mathcal{D} \subseteq \mathcal{C}$ *of bounded linear clique-width which is not* 2*-well-quasi-ordered,*
3. *There exists a regular antichain in* $\mathcal{C}$*,*
4. *There exists a periodic antichain in* $\mathcal{C}$*,*
5. $\mathcal{C}$ *existentially transduces the class of all finite paths.*

We strongly believe that the results of **??** extend to non-hereditary classes of bounded clique-width graphs, but for the sake of clarity and space, we only state and prove them on hereditary ones. On the other hand, let us remark that **??** does not immediately extend to non-hereditary classes, as witnessed by the following example.

▶ **Example 7.** The class $\mathcal{C}$ of complete binary trees has bounded clique-width, is not 2-well-quasi-ordered, but contains no periodic antichain, and every subclass $\mathcal{D} \subseteq \mathcal{C}$ of bounded linear clique-width is 2-well-quasi-ordered. Furthermore, $\mathcal{C}$ existentially transduces the class of all finite paths.

Proof Sketch. It is well known that the class of complete binary trees has bounded clique-width. Furthermore, it is not 2-well-quasi-ordered, since one can colour the leaves and the root of the complete binary trees to form an infinite antichain. On the other hand, any subclass $\mathcal{D} \subseteq \mathcal{C}$ of bounded linear clique-width must be finite: indeed, complete binary trees have unbounded linear clique-width. Finally, because periodic antichains have bounded linear clique-width (see **??**) there cannot be any periodic antichain in $\mathcal{C}$. The existential transduction of all finite paths from $\mathcal{C}$ is obtained by existentially transducing all induced subgraphs (see **??**), which in particular contains all finite paths. ◀

Outline

?? introduces the necessary background on well-quasi-orders, graphs, MSO interpretations and first-order transductions. **??** shows how to reduce the study of graph classes obtained by MSO interpretations from trees to the study of trees labelled over finite monoids. This section also shows how to leverage forward Ramseyan splits to obtain well-behaved tree representations of graphs that are well-quasi-ordered (as trees) with respect to the so-called *gap embedding* relation. **??** identifies combinatorial obstructions to being 2-well-quasi-ordered in images of interpretations from trees, and is the core combinatorial part of the paper. **??** brings together the previous results to prove our **??** and **??**. We also discuss the equivalence between the first two items of **??**. The remaining items of **??** are proven equivalent in **??**. Finally, in **??** we discuss some perspectives and open problems.

2 Preliminaries

We assume the reader to be familiar with basic notions of logic on finite relational structures, such as monadic second-order logic (MSO), first-order logic and quantifier-free formulae.

Graphs A graph G consists of a finite set $V(G)$ of vertices and a set $E(G) \subseteq V(G)^2$ of edges. A graph is *undirected* if $(u, v) \in E$ implies $(v, u) \in E$ for all $u, v \in V$, and directed otherwise. In this paper we will focus on finite undirected graphs without self-loops. Given a set X, an *X-labelled graph* is a graph equipped with a function $\ell \colon V(G) \to X$ that assigns a label to each vertex. Given a class $\mathcal{C}$ of finite undirected graphs, write $\mathsf{Label}_X(\mathcal{C})$ for the class of freely X-labelled graphs in $\mathcal{C}$, i.e., the class of all X-labelled graphs (G, ℓ) such that $G \in \mathcal{C}$.

Whenever $(X, \leq)$ is a quasi-order, we say that a graph G *embeds* into a graph H if there exists a *monomorphism* from G to H, i.e., an injective function $f \colon V(G) \to V(H)$ such that for all $u, v \in V(G)$, $(u, v) \in E(G)$ if and only if $(f(u), f(v)) \in E(H)$, and such that for all $v \in V(G)$, we have $\ell_G(v) \leq \ell_H(f(v))$. We write $G \subseteq_i H$ to denote that G embeds into H, which we also call the *induced subgraph relation*. When no order is specified on a set X, the order relation is assumed to be equality, i.e. $(X, =)$.

A class of graphs $\mathcal{C}$ is a *hereditary class* if it is closed under taking induced subgraphs, i.e. for $G \in \mathcal{C}$, if $H \subseteq_i G$, then $H \in \mathcal{C}$. The *hereditary closure* of a class of graphs $\mathcal{C}$ is the smallest hereditary class containing $\mathcal{C}$.

An *ordered graph* is a finite undirected graph G equipped with a linear order $\leq_G$ on its vertices. The notion of induced subgraph relation extends to ordered graphs by requiring that the embedding $f \colon V(G) \to V(H)$ preserves the order, i.e., for all $u, v \in V(G)$, $u \leq_G v$ if and only if $f(u) \leq_H f(v)$.

Well-quasi-orderings A quasi-ordered set $(X, \leq)$ is a set X with a reflexive and transitive relation $\leq$. A sequence $(x_i)_{n \in \mathbb{N}}$ of elements in X is *good* if there exist $i < j$ such that $x_i \leq x_j$. A quasi-ordered set is *well-quasi-ordered* (*wqo*) if every infinite sequence is good. A *bad sequence* is an infinite sequence that is not good. A sequence of pairwise incomparable elements is called an *antichain*.

As mentioned in the introduction, several notions of well-quasi-ordering can be defined for a given class of finite undirected graphs $\mathcal{C}$. Let us briefly state them: $\mathcal{C}$ is *well-quasi-ordered* if the class $(\mathcal{C}, \subseteq_i)$ is well-quasi-ordered, $\mathcal{C}$ is *k-well-quasi-ordered* for some $k \in \mathbb{N}$, if for every set $(X, =)$ of size k, the class $(\mathsf{Label}_X(\mathcal{C}), \subseteq_i)$ is well-quasi-ordered, and $\mathcal{C}$ is *$\forall$-well-quasi-ordered* if for every well-quasi-order $(X, \leq)$, the class $(\mathsf{Label}_X(\mathcal{C}), \subseteq_i)$ is well-quasi-ordered.

Finally, a class of graphs is *orderably well-quasi-ordered* (resp. *orderably k-well-quasi-ordered*, resp. *orderably $\forall$-well-quasi-ordered*) if there is a way to equip each graph G in $\mathcal{C}$ with a linear order $\leq_G$ on its vertices such that the resulting class of ordered graphs is well-quasi-ordered (resp. k-well-quasi-ordered, resp. $\forall$-well-quasi-ordered).

Trees In this paper, *trees* are finite, binary, and ordered (they have *left* and *right children*). They are understood as finite relational structures over the signature $\sigma = (\sqsubseteq, \preceq)$ where $\sqsubseteq$ is the *ancestor relation*, and $\preceq$ is the *sibling order* on the nodes of the tree. In a tree $\mathcal{T}$, the *root* is denoted by $\mathsf{root}(\mathcal{T})$, with $\mathcal{T}$ being left implicit when clear from context. The set of *leaves* is denoted by $\mathsf{Leaves}(\mathcal{T})$. Given two nodes x and y in a tree T, their *least common ancestor* is denoted by $\mathsf{lca}(x, y)$. We will often use the *parent relation*, which states that y is the unique immediate ancestor of x.

As in the case of graphs, we will consider vertex-labelled trees where labels are taken from a well-quasi-order $(X, \leq)$. We will also consider edge-labelled trees, typically using a finite set of labels Σ. The class of vertex-labelled and edge-labelled trees is denoted by $\mathsf{Trees}_{\Sigma}^{X}$. The sets Σ or X are omitted when unused (i.e., Trees denotes the class of unlabelled trees). Given two nodes x, y such that x is an ancestor of y, the notation $[x : y]_{\mathcal{T}}$ denotes the *word* in Σ^* obtained by reading the labels of the edges on the path from x to y in $\mathcal{T}$; we leave $\mathcal{T}$ implicit when clear from context.

Bounded Clique-Width An MSO-*interpretation* $\mathcal{I}$ from Trees_{Σ} to finite undirected graphs is given by a tuple of MSO formulae that specify how to, respectively, select a subset of the leaves of the tree to be the vertices of the graph ($\phi_{\mathsf{univ}}(x)$), define the edges of the graph ($\phi_{\mathsf{edge}}(x, y)$), and define the domain of the interpretation (ϕ_{dom}). An interpretation $\mathcal{I}$ is called *simple* if $\phi_{\mathsf{univ}}(x)$ and ϕ_{dom} are tautologies.

▶ **Example 8.** An MSO-interpretation $\mathcal{I}$ from Trees to finite undirected graphs that produces all paths can be defined as follows: ϕ_{dom} is the formula that holds on linear trees (i.e., trees where every node has at most one non-leaf child), $\phi_{\mathsf{univ}}(x)$ is the formula that is true if and only if x is a leaf and its sibling is not a leaf, $\phi_{\mathsf{edge}}(x, y)$ is the formula that is true if and only if x and y are at distance exactly 3.

A class of graphs has *bounded clique-width* if it is included in the image of some MSO-interpretation from finite trees to finite undirected graphs [?]. It has *bounded linear clique-width* if it is included in the image of some MSO-interpretation from linear trees to finite undirected graphs. Equivalently, a class of graphs has bounded linear clique-width if it is included in the image of some simple MSO-interpretation from finite words to finite undirected graphs.

Transductions from graphs to graphs Another notion of logical transformation relevant in this paper is that of existential transduction and existential interpretation that were mentioned in **????**, and are defined in a similar way as MSO-interpretations.

An *existential interpretation* from (labelled) graphs to graphs is given by an existential first-order formula $\phi_{\mathrm{univ}}(x)$ that selects the vertices of the interpreted graph, an existential first-order formula $\phi_{\mathrm{edge}}(x, y)$ that defines the edges of the interpreted graph, and an existential first-order formula ϕ_{dom} that selects the graphs of $\mathcal{C}$ on which the interpretation is defined. The semantics is that for every graph $G \in \mathcal{C}$ satisfying ϕ_{dom}, the interpreted graph $\mathcal{I}(G)$ has as vertex set the vertices of G satisfying ϕ_{univ}, and has an edge between two vertices (u, v) if and only if G satisfies $\phi_{\mathrm{edge}}(u, v)$ (which is assumed to be symmetric and irreflexive on the selected vertices). A class $\mathcal{C}$ *existentially interprets* a class $\mathcal{D}$ if there exists an existential interpretation $\mathcal{I}$ from $\mathcal{C}$ to graphs such that $\mathcal{D} \subseteq \mathcal{I}(\mathcal{C})$. We say that $\mathcal{C}$ *existentially transduces* $\mathcal{D}$ if there exists a finite labelling set Σ such that $\mathsf{Label}_{\Sigma}(\mathcal{C})$ existentially interprets $\mathcal{D}$.

▶ **Example 9.** Let $\mathcal{C}$ be a class of graphs. Then, $\mathcal{C}$ existentially transduces the hereditary closure of $\mathcal{C}$, using labels $\Sigma = \{\circ, \bullet\}$, and the interpretation defined as follows: ϕ_{dom} is the tautological formula, $\phi_{\mathrm{univ}}(x)$ is the formula that is true if and only if x is labelled with $\bullet$, $\phi_{\mathrm{edge}}(x, y)$ is the formula that is true if and only if there is an edge between x and y in the original graph.

Let us remark that if a (non-hereditary) class of graphs having bounded clique-width is 3-well-quasi-ordered, then its hereditary closure is 2-well-quasi-ordered, and therefore the class itself is $\forall$-well-quasi-ordered. In particular, one gets a generalisation of **??** (resp. **??**) to non-hereditary classes of graphs of bounded clique-width, up to replacing 2-well-quasi-ordered by 3-well-quasi-ordered. As stated in the introduction, we strongly believe that our results can be extended to non-hereditary classes of graphs of bounded clique-width without this increase in the number of labels, with minimal changes in the proofs.

3 From Interpretations to Nested Trees

As a first step towards proving **??**, we transform a simple MSO-interpretation from trees to graphs into a more combinatorial object that renders the interpretation *quantifier-free*. To that end, we chain two standard transformations from automata theory: we first convert an MSO-interpretation into a so-called monoid interpretation, and then factorise the trees using forward Ramseyan splits (nested trees) [**?**]. Finally, we show that the natural ordering on nested trees (the gap embedding) is *almost* order-preserving for the resulting interpretation from nested trees to graphs.

Let us fix for the rest of the section a finite alphabet Σ and a simple MSO-interpretation $\mathcal{I}$ from Trees_Σ to finite undirected graphs defined by a formula $\varphi(x,y)$ that defines the edges of the graphs.

3.1 From Interpretation to Monoids

It follows from standard arguments that to an MSO formula $\varphi(x,y)$ over Trees_Σ one can associate a finite monoid M and a morphism $\mu\colon \Sigma^* \to M$ such that for any tree T and any pair of leaves $x \preceq y$ in T, whether $T \models \varphi(x,y)$ is entirely determined by the values of $\mu([z : x]_\mathcal{T})$, $\mu([z : y]_\mathcal{T})$ and $\mu([\mathsf{root} : z]_\mathcal{T})$ where $z = \mathsf{lca}(x,y)$ is the least common ancestor of x and y. We refer to the book on Tree Automata Techniques and Applications for a comprehensive overview of the connection between Monadic Second Order Logic on trees, and tree automata [**?**]. In the rest of the section, we fix such a monoid M and morphism μ. To simplify notations, we write $[\![x : y]\!]_\mathcal{T}$ for $\mu([x : y]_\mathcal{T})$, that is, the product of the labels of the edges on the path from x to y in $\mathcal{T}$. If the path is empty (i.e., $x = y$), then $[\![x : y]\!]_\mathcal{T}$ is the identity element of M.

As a consequence, one can collect in a subset $P \subseteq M^3$ all the triples (a,b,c) that correspond to satisfying assignments of the formula $\varphi(x,y)$, as illustrated in **??**. We define a *monoid interpretation* from trees labelled over M to finite undirected graphs to be a tuple (Σ, μ, M, P) where Σ is a finite alphabet, $\mu\colon \Sigma^* \to M$ is a morphism to a finite monoid M, and $P \subseteq M^3$ is a set of triples of elements of M. The interpretation works as follows: given a tree $\mathcal{T}$ whose edges are labelled by elements of Σ, the vertices of the interpreted graph are the leaves of T, and there is an edge between two leaves $x \preceq y$ if and only if the triple $([\![\mathsf{root} : \mathsf{lca}(x,y)]\!]_\mathcal{T}, [\![\mathsf{lca}(x,y) : x]\!]_\mathcal{T}, [\![\mathsf{lca}(x,y) : y]\!]_\mathcal{T})$ belongs to P.

Adding the function $[\![x : y]\!]_T$ to the signature of trees, and assuming that lca is also part of the signature, we can rewrite the formula $\varphi(x,y)$ as a quantifier-free formula as follows:

$$([\![\mathsf{root} : \mathsf{lca}(x,y)]\!]_\mathcal{T}, [\![\mathsf{lca}(x,y) : x]\!]_\mathcal{T}, [\![\mathsf{lca}(x,y) : y]\!]_\mathcal{T}) \in P \tag{1}$$

The main idea driving this paper is that deciding when a class of graphs is $\forall$-well-quasi-ordered reduces to finding a suitable well-quasi-order on trees such that the interpretation $\mathcal{I}$ is *order-preserving* from trees to graphs, leveraging the standard **??**.

▶ **Fact 10** (Folklore). *Let $(X, \leq_X)$ and $(Y, \leq_Y)$ be two quasi-orders, and $f\colon X \to Y$ be a surjective order-preserving map. If $(X, \leq_X)$ is well-quasi-ordered, then $(Y, \leq_Y)$ is well-quasi-ordered.*

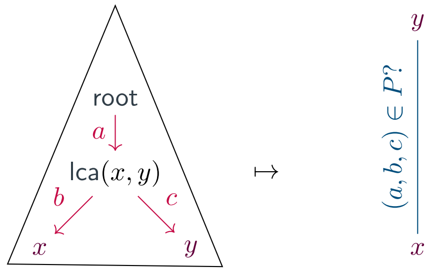

Figure 2 The interpretation of a tree using a monoid M and an accepting part $P \subseteq M^3$.

Similarly, if $f : X \to Y$ is an order reflection, that is, for every x_1, x_2 in X, $f(x_1) \leq_Y f(x_2)$ implies $x_1 \leq_X x_2$, and $(Y, \leq_Y)$ is well-quasi-ordered, then $(X, \leq_X)$ is well-quasi-ordered.

One candidate ordering on trees is the *composition ordering*. Given two trees $\mathcal{T}_1$ and $\mathcal{T}_2$ edge-labelled over a monoid M, we say that $\mathcal{T}_1$ is less than $\mathcal{T}_2$ in the composition ordering, written $\mathcal{T}_1 \preceq_{\mathsf{cmp}} \mathcal{T}_2$, if there exists a map $h : \mathcal{T}_1 \to \mathcal{T}_2$ that maps leaves to leaves and respects lca, $\preceq$, and $[\![\cdot : \cdot]\!]$. This ordering naturally extends to node-labelled trees by requiring that for every node x in $\mathcal{T}_1$, the label of x is less than or equal to the label of $h(x)$ in $\mathcal{T}_2$. It is straightforward to check that the interpretation $\mathcal{I}$ is order-preserving from trees equipped with the composition ordering to graphs equipped with the induced subgraph ordering. Hence, to show that the image of $\mathcal{I}$ is $\forall$-well-quasi-ordered, it suffices to prove that the class of trees is well-quasi-ordered under the composition ordering when labelled using a well-quasi-ordered set $(X, \leq_X)$.

Historically, this approach has been used to prove that some classes of graphs are $\forall$-well-quasi-ordered; see for instance the results of [**?**, **?**]. Unfortunately, the composition ordering on trees is often much stricter than the induced subgraph ordering. It was observed that the composition ordering on trees is well-quasi-ordered if and only if *for every* $P \subseteq M^3$, the class of graphs obtained from trees by considering only the triples in P is $\forall$-well-quasi-ordered [**?**, Theorem 24]. To understand what happens for a specific choice of P, we will develop a finer understanding of the trees themselves.

3.2 Forward Ramseyan Splits

The combinatorial ingredient that allows us to further decompose the trees comes from an adaptation of the classical Simon Factorisation Theorem for semigroups [**?**], adapted to trees by Colcombet [**?**]. The rest of this section explains how this factorisation works and how it can be used to define a more suitable ordering on trees.

A *split of height N* of a tree $\mathcal{T}$ is a mapping $\mathfrak{s}$ from the nodes of $\mathcal{T}$ to $\{1, \ldots, N\}$. Given a split $\mathfrak{s}$ and two nodes $x \sqsubseteq_{\mathcal{T}} y$, we define $\mathfrak{s}(x : y)$ to be the minimal value of $\mathfrak{s}(z)$ for $x \sqsubset_{\mathcal{T}} z \sqsubset_{\mathcal{T}} y$, and $N + 1$ otherwise. In particular, if $x = y$ or y is a child of x, then $\mathfrak{s}(x : y) = N + 1$.

Let us fix a branch B of $\mathcal{T}$, that is, a maximal set of nodes of $\mathcal{T}$ that are pairwise comparable for $\sqsubseteq$. We define an equivalence relation on the notes of B as follows: two nodes x and y of B are in the same class if $\mathfrak{s}(x) = \mathfrak{s}(y) = k$ and $\mathfrak{s}(x : y) \geq k$. A split thus induces a hierarchical structure on each branch based on its k-neighbourhoods. We illustrate the situation in **??**. We refer to the value of $\mathfrak{s}(x)$ as the *split depth* of the node x.

A split $\mathfrak{s}$ is *forward Ramseyan* if for every $k \in \{1, \ldots, N\}$, every branch B of the tree,

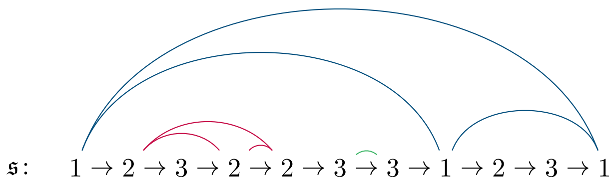

Figure 3 A split $\mathfrak{s}$ on a branch of a tree. The arcs in blue connect 1-neighbours, the arcs in red connect 2-neighbours, and the arcs in green connect 3-neighbours.

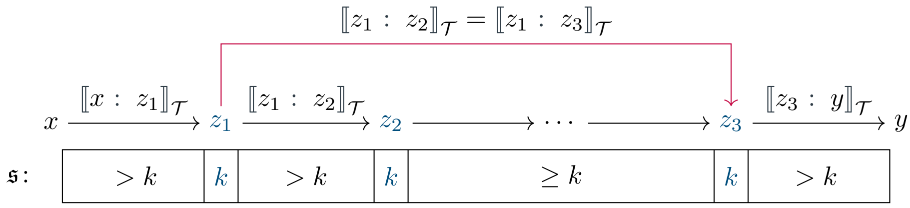

Figure 4 Fast computation of the value $[\![x : y]\!]_{\mathcal{T}}$ provided a forward Ramseyan split, using the fact that x and y are independent at level k.

and every x, y, x', y' in the same k-neighbourhood of B, with $x \sqsubset_{\mathcal{T}} y$ and $x' \sqsubset_{\mathcal{T}} y'$, we have:

$$[\![x : y]\!]_{\mathcal{T}} = [\![x : y]\!]_{\mathcal{T}} \cdot [\![x' : y']\!]_{\mathcal{T}} \quad . \tag{2}$$

In particular, $[\![x : y]\!]_{\mathcal{T}}$ is always an *idempotent*, that is, it satisfies $e \cdot e = e$. However, $[\![x : y]\!]_{\mathcal{T}}$ and $[\![x' : y']\!]_{\mathcal{T}}$ may be different idempotents.

Let us illustrate how one can use forward Ramseyan splits to compute efficiently the value of $[\![x : y]\!]_{\mathcal{T}}$ for any pair of nodes $x \sqsubset_{\mathcal{T}} y$. We say that x and y are *independent at level k* if $\mathfrak{s}(x : y) = k$ and there exist three nodes z_1, z_2, z_3 such that $x \sqsubseteq_{\mathcal{T}} z_1 \sqsubset_{\mathcal{T}} z_2 \sqsubseteq_{\mathcal{T}} z_3 \sqsubseteq_{\mathcal{T}} y$, $\mathfrak{s}(z_1) = \mathfrak{s}(z_2) = \mathfrak{s}(z_3) = k$, and $\mathfrak{s}(x\colon z_1) > k$, $\mathfrak{s}(z_1\colon z_2) > k$, and $\mathfrak{s}(z_3\colon y) > k$. In this case, we can exploit forward Ramseyan splits to compute $[\![x : y]\!]_{\mathcal{T}}$ without inspecting the section between z_2 and z_3, leading to a fast (and first-order definable) computation. We refer to [**?**, Lemma 3] for further details.

The main theorem of [**?**] states that for every finite monoid M there exists a finite depth N such that for every tree $\mathcal{T}$ edge-labelled using M, there is a forward Ramseyan split of height N for $\mathcal{T}$. We can therefore assume that our trees are always equipped with a forward Ramseyan split of height N.

We now state the main remark that guides the rest of the paper: given a branch of a tree equipped with a forward Ramseyan split as in **??**, and assuming that $z_2 \neq z_3$, the value of $[\![x : y]\!]_{\mathcal{T}}$ does not change when inserting new nodes in the section between z_2 and z_3, provided these new nodes are assigned a split value at least k and the resulting tree remains forward Ramseyan. A fortiori, whether there is an edge between the leaves x and y in the resulting graph is not affected by such insertions. This observation guides our definition of a suitable ordering on trees in the upcoming **??**.

3.3 Marked Nested Trees

In this section we show how the notion of forward Ramseyan split brings us closer to defining a well-quasi-order on trees that suits our purpose. A first step is to note that there already exists a notion of embedding that respects forward Ramseyan splits, namely gap-embeddings, which endow trees with a well-quasi-order [?].

▶ **Definition 11** ([?, Definition 3.3]). *A gap-embedding between trees $\mathcal{T}_1, \mathcal{T}_2$ endowed with forward Ramseyan splits $\mathfrak{s}_1, \mathfrak{s}_2$ is a mapping $h\colon \mathcal{T}_1 \to \mathcal{T}_2$ that*

1. *is a tree embedding: respects the ancestor relation $\sqsubseteq$, least common ancestors* lca*, and sibling ordering $\preceq$,*
2. *satisfies the root gap property: $\mathfrak{s}_2(r_2\colon h(r_1)) \geq \mathfrak{s}_1(r_1)$ where r_1, r_2 are the roots of $\mathcal{T}_1, \mathcal{T}_2$,*
3. *satisfies the edge gap property: for every edge $x \sqsubset_{\mathcal{T}} y$ in $\mathcal{T}_1$, $\mathfrak{s}_2(h(x)\colon h(y)) \geq \mathfrak{s}_1(y)$.*

If the trees are node-labelled by a set $(X, \leq_X)$, we further require that for every node x in $\mathcal{T}_1$, the label of x is less than or equal to the label of $h(x)$ in $\mathcal{T}_2$.

In **??** we illustrate why gap-embeddings are well suited to our purpose: they respect the gaps induced by forward Ramseyan splits. Another key property of the gap-embedding ordering is that it is a well-quasi-ordering, as recalled in **??**.

▶ **Fact 12.** *Let $\mathcal{T}_1, \mathcal{T}_2$ be two trees and $\mathfrak{s}_1, \mathfrak{s}_2$ be two forward Ramseyan splits of height N of $\mathcal{T}_1, \mathcal{T}_2$ Assume that $h\colon \mathcal{T}_1 \to \mathcal{T}_2$ is a gap-embedding from $(\mathcal{T}_1, \mathfrak{s}_1)$ to $(\mathcal{T}_2, \mathfrak{s}_2)$. Then, for every u, v in $\mathcal{T}_1$, if $\mathfrak{s}_1(u : v) > k$ and $\mathfrak{s}_1(v) = k$ for some $k \in \{1, \ldots, N\}$, then $\mathfrak{s}_2(h(u) : h(v)) \geq k$.*

▶ **Theorem 13** ([?, Theorem 3.1, Main Theorem]). *The class of trees equipped with forward Ramseyan splits is well-quasi-ordered under the gap-embedding ordering, even under the assumption that nodes are labelled with a well-quasi-ordered set $(X, \leq_X)$.*

We now devise a modified version of the gap-embedding ordering to make the interpretation $\mathcal{I}$ order-preserving from trees to graphs. In particular, this modification will take into account the edge-labelling of the trees using the monoid M, that are not considered in the gap-embedding ordering.

▶ **Definition 14.** *A marked nested tree is a tuple $(\mathcal{T}, \mathfrak{s}, \rho)$ where $\mathcal{T}$ is a tree whose edges are labelled over a finite monoid M, $\mathfrak{s}$ is a forward Ramseyan split of height N of $\mathcal{T}$, and ρ is a function from the nodes of $\mathcal{T}$ to $\{\mathbf{M}, \mathbf{S}, \mathbf{D}\}$, respectively called marked, separating, and dummy nodes.*

The definition of 'well-marked' needs tidying up. First, it's currently quantified over every x, z_1 and y, but then we have the additional condition 'if x and y are marked nodes', while later it says that also $\rho(z_1) = \mathbf{M}$. What is being quantified over to check whether this definition holds? Additionally, '$\mathfrak{s}(z_1) = k$' is said twice, as is '$\mathfrak{s}(x : z_1) > k$', and there is the perennial issue of interpreting what to do when, e.g., $z_2 = z_3$ or $z_3 = y$.

A marked nested tree $(\mathcal{T}, \mathfrak{s}, \rho)$ is well-marked if the marked nodes contain the root of $\mathcal{T}$, are closed under the least common ancestor operation, and furthermore satisfy that for every $x \sqsubset_{\mathcal{T}} z_1 \sqsubset_{\mathcal{T}} y$ with $\mathfrak{s}(x\colon z_1) > k$, $\mathfrak{s}(z_1) = k$, and $\mathfrak{s}(z_1\colon y) \geq k$ for some $k \in \{1, \ldots, N\}$, if x and y are marked nodes, then so is Z_1 and there exist z_2, z_3 such that $z_1 \sqsubset_{\mathcal{T}} z_2 \sqsubseteq_{\mathcal{T}} z_3 \sqsubseteq_{\mathcal{T}} y$, $\mathfrak{s}(z_2) = \mathfrak{s}(z_3) = k$, $\mathfrak{s}(z_1\colon z_2) > k$, $\mathfrak{s}(z_2\colon z_3) \geq k$, and $\mathfrak{s}(z_3\colon y) \geq k$, with $\rho(z_2) \in \{\mathbf{M}, \mathbf{S}\}$. A graphical illustration of the condition is provided below:

Mael: Suggestion above, but the definition is still extremely convoluted. It's always disturbing the $z_1 \sqsubset_{\mathcal{T}} z_2 \sqsubseteq_{\mathcal{T}} z_3 \sqsubseteq_{\mathcal{T}} y$.

$$\begin{array}{ccccccccc}
\mathbf{M} & & \mathbf{M} & & \in \{\mathbf{M}, \mathbf{S}\} & & \in \{\mathbf{M}, \mathbf{S}, \mathbf{D}\} & & \mathbf{M} \\
x & \xrightarrow[> k]{} & z_1 & \xrightarrow[> k]{} & z_2 & \xrightarrow[\geq k]{} & z_3 & \xrightarrow[\geq k]{} & y \\
 & & k & & k & & k & &
\end{array}$$

▶ **Definition 15.** *Given two marked nested trees* $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1)$ *and* $(\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$*, we say that* $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1)$ *gap-embeds into* $(\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$*, written* $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1) \leq_{\text{gap}} (\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$*, if there exists a mapping* $h \colon \mathcal{T}_1 \to \mathcal{T}_2$ *that*

- ⋆ *is a gap embedding from* $(\mathcal{T}_1, \mathfrak{s}_1)$ *to* $(\mathcal{T}_2, \mathfrak{s}_2)$*,*

4. *sends the root of* $\mathcal{T}_1$ *to the root of* $\mathcal{T}_2$*,*
5. *maps leaves to leaves,*
6. *respects the marking:* $\rho_1 = \rho_2 \circ h$*,*
7. *respects local products: if* y *is the left (resp. right) child of* x *in* $\mathcal{T}_1$*, and* y' *is the left (resp. right) child of* $h(x)$ *in* $\mathcal{T}_2$*, then* $[\![x : y]\!]_{\mathcal{T}_1} = [\![h(x) : y']\!]_{\mathcal{T}_2}$*.*
8. *respects neighbourhood products: for every* $k \in \{1, \ldots, N\}$*, and every node* $x \in \mathcal{T}_1$*, if* z *is the closest ancestor of* x *such that* $\mathfrak{s}_1(z) = k$ *and* z' *is the closest ancestor of* $h(x)$ *such that* $\mathfrak{s}_2(z') = k$*, then* $[\![z : x]\!]_{\mathcal{T}_1} = [\![z' : h(x)]\!]_{\mathcal{T}_2}$*.*
9. *is gluing on non-dummy nodes: if* $\rho_1(y) \in \{\mathbf{M}, \mathbf{S}\}$*, and* x *is the parent of* y *in* $\mathcal{T}_1$*, then* $h(x)$ *is the parent of* $h(y)$ *in* $\mathcal{T}_2$*.*

Unlike the gap-embedding ordering of **??**, the marked gap-embedding ordering of **??** is not a well-quasi-order because of the gluing condition in **??**. We can create an infinite antichain by considering the infinite sequence of trees $(\mathcal{T}_n, \mathfrak{s}_n, \rho_n)$ where $\mathcal{T}_n$ is a branch of length $n+1$, $\mathfrak{s}_n$ assigns the value 1 to every node of $\mathcal{T}_n$, and ρ_n marks all the nodes of $\mathcal{T}_n$ with $\mathbf{M}$. It is straightforward to check that for every $n < m$, $(\mathcal{T}_n, \mathfrak{s}_n, \rho_n)$ does not gap-embed into $(\mathcal{T}_m, \mathfrak{s}_m, \rho_m)$.

To recover the well-quasi-order property, we will restrict our attention to trees that make a limited use of the gluing property. An *L-bounded marked nested tree* is a marked nested tree such that any path of non-dummy nodes has length at most L. We prove in **??** that limiting the use of gluing restores the well-quasi-order property. Furthermore, we show in **??** that gap-embeddings between well-marked nested trees encode the composition ordering on trees when restricted to marked nodes. The combination of these two results in **??** is the main combinatorial ingredient of this section.

▶ **Theorem 16.** *For every finite* L*, the class of* L*-bounded well-marked nested trees labelled over a well-quasi-ordered set* $(X, \leq_X)$ *is well-quasi-ordered under the gap-embedding ordering.*

▶ **Lemma 17.** *Let* $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1)$ *and* $(\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$ *be two marked nested trees such that* $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1) \leq_{\text{gap}} (\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$ *via some mapping* h*. For every pair of marked nodes* $x \sqsubset_{\mathcal{T}_1} y$ *in* $\mathcal{T}_1$*, we have*

$$[\![x : y]\!]_{\mathcal{T}_1} = [\![h(x) : h(y)]\!]_{\mathcal{T}_2} \quad .$$

Proof. We proceed by induction on the value of $\mathfrak{s}_1(x \colon y)$. If $\mathfrak{s}_1(x \colon y) = N+1$, then y is a child of x. Because y is a marked node, then by **??** we know that $h(y)$ is a child of $h(x)$. Therefore, by **??**, we have $[\![x : y]\!]_{\mathcal{T}_1} = [\![h(x) : h(y)]\!]_{\mathcal{T}_2}$.

Otherwise, let $k = \mathfrak{s}_1(x : y) \leq N$. Since $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1)$ is well-marked, there exists z_1, z_2, z_3 such that $x \sqsubset_{\mathcal{T}_1} z_1 \sqsubset_{\mathcal{T}_1} z_2 \sqsubseteq_{\mathcal{T}_1} z_3 \sqsubseteq_{\mathcal{T}_1} y$, $\mathfrak{s}_1(z_1) = \mathfrak{s}_1(z_2) = \mathfrak{s}_1(z_3) = k$, and $\mathfrak{s}_1(x \colon z_1) > k$, $\mathfrak{s}_1(z_1 \colon z_2) > k$, $\mathfrak{s}_1(z_2 \colon z_3) \geq k$, and $\mathfrak{s}_1(z_3 \colon y) > k$, with $\rho_1(z_1) = \mathbf{M}$, and $\rho_1(z_2) \in \{\mathbf{M}, \mathbf{S}\}$. This is depicted in the following figure:

$$x \xrightarrow[> k]{[\![x : z_1]\!]_{\mathcal{T}_1}} \underset{k}{z_1} \xrightarrow[> k]{[\![z_1 : z_2]\!]_{\mathcal{T}_1}} \underset{k}{z_2} \xrightarrow[\geq k]{[\![z_2 : z_3]\!]_{\mathcal{T}_1}} \underset{k}{z_3} \xrightarrow[> k]{[\![z_3 : y]\!]_{\mathcal{T}_1}} y$$

By applying the gap-embedding h, we have the following configuration inside $\mathcal{T}_2$, where we used the notations $a = [\![h(x) : h(z_1)]\!]_{\mathcal{T}_2}$, $b = [\![h(z_1) : h(z_2)]\!]_{\mathcal{T}_2}$, $c = [\![h(z_2) : h(z_3)]\!]_{\mathcal{T}_2}$, and $d = [\![h(z_3) : h(y)]\!]_{\mathcal{T}_2}$.

$$\begin{array}{ccccccccc} h(x) & \xrightarrow[>k]{a} & h(z_1) & \xrightarrow[>k]{b} & h(z_2) & \xrightarrow[\geq k]{c} & h(z_3) & \xrightarrow[>k]{d} & h(y) \\ & & k & & k & & k & & \end{array}$$

Let us briefly argue the inequalities on the splits on the above paths. The non-strict inequalities follow from **??**. Furthermore, if v is a marked node or a separating node, then we can strengthen the inequality to a strict one, i.e., $\mathfrak{s}(h(u) \colon h(v)) > k$ even when $\mathfrak{s}(v) \neq k$. This is because of **??**, which ensures that the parent of $h(v)$ is the image of the parent of v, which has a split value strictly greater than k, and we conclude by the previous argument.[3]

Using these facts we conclude that:

$$\begin{aligned}
\mathfrak{s}_2(h(x) \colon h(z_1)) &> k && \text{because } z_1 \text{ is marked,}\\
\mathfrak{s}_2(h(z_1) \colon h(z_2)) &> k && \text{because } z_2 \text{ is marked or separating,}\\
\mathfrak{s}_2(h(z_2) \colon h(z_3)) &\geq k && \text{by ??,}\\
\mathfrak{s}_2(h(z_3) \colon h(y)) &> k && \text{because } y \text{ is marked,}
\end{aligned}$$

Now, we can compute the values of a, b, and d:

$$\begin{aligned}
[\![x : z_1]\!]_{\mathcal{T}_1} &= a && \text{by induction hypothesis,}\\
[\![z_1 : z_2]\!]_{\mathcal{T}_1} &= b && \text{by ??,}\\
[\![z_3 : y]\!]_{\mathcal{T}_1} &= d && \text{by ??.}
\end{aligned}$$

Finally, because $h(x)$ and $h(y)$ are independent at level k, and so are x and y, we have by **??** that

$$\begin{aligned}
[\![x : y]\!]_{\mathcal{T}_1} &= [\![x : z_1]\!]_{\mathcal{T}_1} \cdot [\![z_1 : z_2]\!]_{\mathcal{T}_1} \cdot [\![z_2 : z_3]\!]_{\mathcal{T}_1} \cdot [\![z_3 : y]\!]_{\mathcal{T}_1}\\
&= [\![x : z_1]\!]_{\mathcal{T}_1} \cdot [\![z_1 : z_2]\!]_{\mathcal{T}_1} \cdot [\![z_3 : y]\!]_{\mathcal{T}_1}\\
&= a \cdot b \cdot d\\
&= a \cdot b \cdot c \cdot d\\
&= [\![h(x) : h(y)]\!]_{\mathcal{T}_2} \quad .
\end{aligned}$$ ◀

▶ **Corollary 18.** *Let $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1)$ and $(\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$ be two well-marked nested trees such that $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1) \leq_{\text{gap}} (\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$. Then, the graph interpreted from $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1)$ is an induced subgraph of the graph interpreted from $(\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$ when considering only marked leaves as vertices.*

Proof. Follows directly from **??** and the definition of the monoid interpretation using the set $P \subseteq M^3$ (see **??**): edges between two marked leaves x and y only depend on the values of $[\![\mathsf{root} : \mathsf{lca}(x, y)]\!]_{\mathcal{T}}$, $[\![\mathsf{lca}(x, y) : x]\!]_{\mathcal{T}}$, and $[\![\mathsf{lca}(x, y) : y]\!]_{\mathcal{T}}$, all of which are marked nodes because the trees are well-marked. ◀

[3] The case where there are no intermediate nodes between u and v happens if v is the child of u, in which case the property is given directly by the gluing property.

4 Bad Patterns in Images of Interpretations

In this section we fix a monoid interpretation $\mathcal{I} = (\Sigma, \mu, M, P)$ from trees edge-labelled over a finite alphabet Σ to finite undirected graphs. We furthermore assume that all trees we consider are equipped with a forward Ramseyan split $\mathfrak{s}$ over the finite monoid M. Our goal is to isolate some combinatorial obstructions to being 2-well-quasi-ordered in classes of graphs that are images of monoid interpretations from trees. The key idea of this section is to start from a tree $\mathcal{T}$ equipped with a forward Ramseyan split $\mathfrak{s}$, and try to understand under which conditions one can turn this tree into a L-bounded marked nested tree representing the same graph, allowing us to leverage **??** to conclude that the class of graphs is $\forall$-well-quasi-ordered.

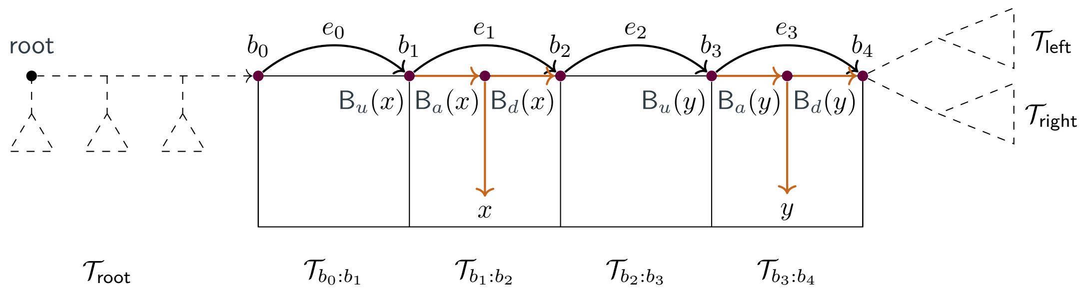

Figure 5 Partitioning a branch of a tree $\mathcal{T}$ using a bough B. Here, the nodes b_i for $0 \le i \le 4$ are k-neighbours forming the backbone B. The letter e_i denotes idempotent monoid element $[\![b_i : b_{i+1}]\!]_{\mathcal{T}}$. In dashed, we represented the context $C[\square]$ such that $\mathcal{T} = C[B]$. To compute the presence of an edge between x and y in the resulting graph, it is sufficient to know the values of $[\![b_2 : b_3]\!]_{\mathcal{T}}$, and the respective bough types of x and y in their respective bough blocks $\mathcal{T}_{b_1:b_2}$ and $\mathcal{T}_{b_3:b_4}$.

distinct ?

▶ **Definition 19.** *Let $\mathcal{T}$ be a tree and $\mathfrak{s}$ be a forward Ramseyan split of height N. Let $n \ge 1$, and let $b_0, b_1, \ldots, b_n$ be nodes on a branch of $\mathcal{T}$ such that $b_0 \sqsubseteq b_1 \sqsubseteq \cdots \sqsubseteq b_n$, and for every $0 \le i < n$, $\mathfrak{s}(b_i) = k$ and $\mathfrak{s}(b_i : b_{i+1}) > k$. A bough of level k is the subtree of $\mathcal{T}$ induced by all nodes x such that $b_0 \sqsubseteq x$ and $\neg(b_n \sqsubseteq x)$, plus b_n itself. We say that the dimension of the bough B is n, and that $b_0, b_1, \ldots, b_n$ form the backbone of B.*

Mael: I think the current definition of Bough is fine, and that for a given bough of level k (defined from a backbone), the backbone is uniquely defined. But we might add that whenever we define a bough B we think of it as a subtree with a highlighted root-leaf branch

Let B be a bough of level k in $\mathcal{T}$ and let $b_0, b_1, \ldots, b_n$ be its backbone. Given indices $i < j$, we define the $\mathcal{T}_{b_i:b_j}$ as the set of nodes x in $\mathcal{T}$ such that $b_i \sqsubseteq x$ and $\neg(b_j \sqsubseteq x)$. When $j = i + 1$, this set of nodes is called a *block*. We also refer to **??** for an illustration of the resulting partition of the tree $\mathcal{T}$ with respect to a given bough B.

For every leaf x of $\mathcal{T}$ that is in $\mathcal{T}_{b_0:b_n}$, we define $\mathsf{B}_a(x) \triangleq \mathsf{lca}(x, b_n)$. Moreover, we define $\mathsf{B}_u(x)$ (resp. $\mathsf{B}_d(x)$) to be the maximal (resp. minimal) element of the backbone $b_0, b_1, \ldots, b_n$ such that $\mathsf{B}_u(x) \sqsubseteq \mathsf{B}_a(x) \sqsubseteq \mathsf{B}_d(x)$. Using these reference points, we define the following tuple monoid elements associated to a leaf x in a block of B:

$$([\![\mathsf{B}_a(x) : x]\!]_{\mathcal{T}}, [\![\mathsf{B}_u(x) : \mathsf{B}_a(x)]\!]_{\mathcal{T}}, [\![\mathsf{B}_a(x) : \mathsf{B}_d(x)]\!]_{\mathcal{T}}, [\![\mathsf{root} : \mathsf{B}_u(x)]\!]_{\mathcal{T}}).$$

simplified notation as we didn't use $\mathcal{B}_t(x)$ + changed to ancestor, up, down instead of t,left,right

We call this tuple the *bough type* of the leaf x with respect to the bough B. We again refer to **??** for an illustration of the type of a leaf with respect to a given bough B. There are only finitely many possible bough types because M is finite.

Throughout the paper, we often rely on tree surgery operations consisting of replacing a bough in a tree by another bough. Given a bough B in a tree $\mathcal{T}$ defined using a backbone $b_0, b_1, \ldots, b_n$, we write $\mathcal{T} = C[B]$ to denote that $\mathcal{T}$ is obtained by plugging the bough B into a *context* $C[\square]$. Formally, a *context* $C[\square]$ is given by a tree $\mathcal{T}_{\mathsf{root}}$ with a distinguished leaf $\square$, together with two trees $\mathcal{T}_{\mathsf{left}}$ and $\mathcal{T}_{\mathsf{right}}$, and two monoid elements m_{left} and m_{right}. These elements are represented as dashed lines in **??**. The tree $C[B]$ is then obtained as follows: one first attaches $\mathcal{T}_{\mathsf{left}}$ as the left subtree of b_n with an edge labelled m_{left}, then attaches $\mathcal{T}_{\mathsf{right}}$ as the right subtree of b_n with an edge labelled m_{right}, and finally replaces the distinguished leaf $\square$ in $\mathcal{T}_{\mathsf{root}}$ with the node b_0.

In order to simplify statements, we will say that the *leaves* of a context $C[\square]$ are the leaves of the tree $\mathcal{T}_{\mathsf{root}}$, $\mathcal{T}_{\mathsf{left}}$, and $\mathcal{T}_{\mathsf{right}}$, excluding the distinguished leaf $\square$. As a consequence, the leaves of $C[B]$ are exactly the leaves of $C[\square]$ together with the leaves of the blocks of B. Furthermore, we will often need to talk about the *first idempotent value* of a bough B defined using the backbone $b_0, b_1, \ldots, b_n$, which is the idempotent element $[\![b_0 : b_1]\!]_{\mathcal{T}}$.[4]

Given two boughs B and B' of level k, and a context $C[\square]$, one can consider the trees $C[B]$ and $C[B']$. These trees have edges labelled over M and are equipped with splits induced by the splits on $C[\square]$ and B (resp. B'). We say that B and B' are *compatible* if they share the same first idempotent value and if for every context $C[\square]$ such that the tree $C[B]$ (with split $\mathfrak{s}$) is forward Ramseyan, the tree $C[B']$ (with split $\mathfrak{s}'$) is also forward Ramseyan.

▶ **Definition 20.** *Let B be a bough of level k in a tree $\mathcal{T} = C[B]$. We say that B is a* good bough *if there exists a compatible bough H of level k and a map $h \colon \mathcal{I}(C[B]) \to \mathcal{I}(C[H])$ such that:*

1. *h is an embedding of graphs,*
2. *h is the identity map on leaves of $C[\square]$,*
3. *there exist three consecutive blocks in H whose leaves are left untouched[5] by h.*

A bad bough *is a bough that is not a good bough.*

We claim that the existence of bad boughs of arbitrarily large dimension is the only obstruction to being 2-well-quasi-ordered.

▶ **Lemma 21.** *The following are equivalent:*

1. *The image of $\mathcal{I}$ is 2-well-quasi-ordered.*
2. *The image of $\mathcal{I}$ is $\forall$-well-quasi-ordered.*
3. *There exists a bound on the dimension of bad boughs.*

It is clear that **??** implies **??**. In **??** we will prove that **??** implies **??**, while in **??** we will prove that **??** implies **??**, concluding the proof of **??**.

4.1 Using the Gap Embedding

In this section, we leverage **??** to prove that **??** implies **??** of **??**. The main idea is to use the bound on the dimension of bad boughs to insert “dummy nodes” inside our trees, turning them into marked nested trees. Because we can insert these dummy nodes in a controlled manner, we are able to ensure that the resulting marked nested trees are L-bounded for some $L \in \mathbb{N}$, allowing us to apply **??**.

[4] It is also the element $[\![b_0 : b_n]\!]_{\mathcal{T}}$, thanks to the forward Ramseyan property of $\mathfrak{s}$.

[5] Here we say that a leaf of H is untouched if the corresponding vertex of $\mathcal{I}(C[H])$ is not in the image of h.

▶ **Lemma 22.** *Assume that there exists a bound N_0 on the dimension of bad boughs. Then, for every tree $\mathcal{T}$ equipped with a forward Ramseyan split $\mathfrak{s}$, one can effectively construct a well-marked nested tree $(\mathcal{T}', \mathfrak{s}', \rho)$ that is L-bounded for some $L \in \mathbb{N}$, and such that $\mathcal{I}(\mathcal{T})$ is an induced subgraph of $\mathcal{I}(\mathcal{T}')$, when the latter is restricted to vertices corresponding to marked leaves.*

Proof. We proceed iteratively, starting from the tree $\mathcal{T}$ where every node is set as a marked node. A rewrite step works as follows: given a bough backbone $b_0, b_1, \dots, b_n$ defining a bough B of level k where all nodes of the bough backbone are marked nodes and $n > N_0$, we know that B is a good bough by assumption, and we write $\mathcal{T} = C[B]$.

changed marking as (vertex)-labelling in this paragraph

By definition of good boughs, there exists a compatible bough H and a map $h\colon \mathcal{I}(C[B]) \to \mathcal{I}(C[H])$ satisfying the three conditions of **??**. Let us define $\mathcal{T}' = C[H]$, which is still forward Ramseyan by definition of compatible boughs. Let us now define the vertex-labelling of $\mathcal{T}'$. For nodes of $C[\square]$, we keep the same vertex-labelling as in $\mathcal{T}$. For nodes of H, we identify the three consecutive blocks whose leaves are left untouched by h as $H_{z_i:z_{i+1}}$, $H_{z_{i+1}:z_{i+2}}$, and $H_{z_{i+2}:z_{i+3}}$. We label all these nodes as dummy nodes except z_i and z_{i+3}, which are labelled as marked nodes, and z_{i+1}, which is labelled as a separating node. Finally, all other nodes of H are labelled as marked nodes.

We claim that one can apply these rewrite steps as follows: first, apply them to boughs of minimal split level until exhaustion, and then proceed with the next split level, and so on. To a tree $\mathcal{T}$ and a split level k, one can associate the multiset of dimensions of all boughs of level k with backbone made of marked nodes inside $\mathcal{T}$. Since each rewrite step decreases this multiset for the Dershowitz–Manna ordering this process terminates [**?**].

Let us first observe that every rewrite step preserves the fact that $\mathcal{T}$ is a well-marked nested tree. This is because non-marked nodes are only inserted after a separating node.

Let us now prove that every rewrite step preserves the fact that the graph $\mathcal{I}(\mathcal{T})$ restricted to vertices corresponding to marked leaves is an induced subgraph of $\mathcal{I}(\mathcal{T}')$ restricted to vertices corresponding to marked leaves. This follows directly from the fact that $h\colon \mathcal{I}(C[B]) \to \mathcal{I}(C[H])$ is an embedding of graphs and that h is the identity on leaves of $C[\square]$.

From our construction, it is clear that at every level k, there are at most N_0+1 consecutive marked nodes in the bough backbone of any bough of level k.

Let us now prove that the resulting tree $\mathcal{T}'$ is L-bounded for some $L \in \mathbb{N}$. Assume towards a contradiction that there exists arbitrarily long paths of non-dummy nodes in the resulting tree $\mathcal{T}'$. To a pair of nodes $x \sqsubseteq y$ in such a path, we associate the colour $(\mathfrak{s}'(x), \mathfrak{s}'(x\colon y), \mathfrak{s}'(y))$. By a Ramsey argument, if the path is too long, there exists a sequence $x_0 \sqsubseteq x_1 \sqsubseteq \cdots \sqsubseteq x_{N_0+1}$ of non-dummy nodes in the path such that $\mathfrak{s}'(x_i) = \mathfrak{s}'(x_j)$ and $\mathfrak{s}'(x_i\colon x_j) = \mathfrak{s}'(x_{i'}\colon x_{j'})$ for every $0 \le i < j \le N_0+1$ and $0 \le i' < j' \le N_0+1$. Without loss of generality, we can assume that $\mathfrak{s}'(x_i) < \mathfrak{s}'(x_i\colon x_{i+1})$ for every $0 \le i \le N_0$. Since we only insert separating node prior to a dummy node, we can also assume that all nodes $x_0, x_1, \dots, x_{N_0+1}$ are marked nodes. We have identified a bough backbone $x_0, x_1, \dots, x_{N_0+1}$ of marked nodes defining a bough of level $\mathfrak{s}'(x_0)$, which contradicts the fact that the rewriting procedure terminated. ◀

Proof of ?? implies ?? of ??. Let us assume that there exists a bound on the dimension of bad boughs, and let us consider a well-quasi-ordered set $(Y, \le)$ of labels. Let $(G_i)_{i\in\mathbb{N}}$ be an infinite sequence of graphs in $\mathsf{Label}_Y(\mathcal{I}(\mathsf{Trees}_\Sigma))$. We want to show that there exist two indices $i < j$ such that G_i is a labelled induced subgraph of G_j.

First, every graph G_i can be represented as $\mathcal{I}(\mathcal{T}_i)$ for some tree $\mathcal{T}_i$ equipped with a forward Ramseyan split $\mathfrak{s}_i$. Then, by **??**, there exists a finite $L \in \mathbb{N}$ such that we can construct

L-bounded well-marked nested tree $(\mathcal{T}'_i, \mathfrak{s}'_i, \rho_i)$ such that G_i is an induced subgraph of $\mathcal{I}(\mathcal{T}'_i)$, when the latter is restricted to vertices corresponding to marked leaves. By labelling the marked leaves of $(\mathcal{T}'_i, \mathfrak{s}'_i, \rho_i)$ with $\circ$ if they are not in G_i, and with their original label in Y otherwise, we obtain a sequence of $Y \cup \{\circ\}$ node-labelled well-marked nested trees.

By **??**, there exist two trees $(\mathcal{T}'_i, \mathfrak{s}'_i, \rho'_i)$ and $(\mathcal{T}'_j, \mathfrak{s}'_j, \rho'_j)$ in this sequence such that $(\mathcal{T}_i, \mathfrak{s}_i, \rho_i) \leq_{\text{gap}} (\mathcal{T}_j, \mathfrak{s}_j, \rho_j)$. By **??**, the graph interpreted from $(\mathcal{T}_i, \mathfrak{s}_i, \rho_i)$ is a labelled induced subgraph of the graph interpreted from $(\mathcal{T}'_j, \mathfrak{s}'_j, \rho'_j)$ when both are restricted to marked leaves. We conclude that G_i is a labelled induced subgraph of G_j, and thus $\mathsf{Label}_Y(\mathcal{I}(\mathsf{Trees}_\Sigma))$ is well-quasi-ordered. Therefore, the image of $\mathcal{I}$ is $\forall$-well-quasi-ordered. ◀

4.2 Obstructions

In this section we prove that **??** implies **??** of **??**. We proceed by contraposition and assume that there exist bad boughs of arbitrarily large dimension. We use this to construct an infinite antichain in the image of $\mathcal{I}$ using two colours. To build such an infinite antichain, we extract from an infinite sequence of bad boughs of increasing dimension some incomparable graphs. We first show that all these bad boughs can be assumed to be compatible (see **??** below), and that we can place all these bad boughs in the *same* context $C[\square]$ to witness that they are bad (see **??** below).

▶ **Lemma 23.** *Let k be a level. There are finitely many boughs of level k up to compatibility.*

▶ **Lemma 24.** *There exists a finite set $\mathcal{F}$ of contexts $C[\square]$ such that for every bad bough B of level k, there exists a context $C[\square] \in \mathcal{F}$ such that B is a bad bough in $C[B]$.*

Proof. The proof relies on the fact that the context $C[\square]$ plays a very little role in witnessing that a bough B is bad.

Let B and H be two compatible boughs of level k, and let $C[\square]$ be a context such that $\mathcal{T} = C[B]$ and $\mathcal{T}' = C[H]$ are forward Ramseyan. We want to characterise when there exists an embedding $h: \mathcal{I}(\mathcal{T}) \to \mathcal{I}(\mathcal{T}')$ witnessing that B is a good bough in $C[B]$ with respect to H.

First, we only need to consider edges between leaves of B or edges between leaves of B and leaves of $C[\square]$. Indeed, the map h acts as the identity on leaves of $C[\square]$, so edges are preserved there.

The presence of edges between leaves of B depends only on their respective bough types and the value of $[\![\mathsf{root} : \square]\!]_{\mathcal{T}_{\mathsf{root}}}$. Consequently, any two contexts sharing the same value for the path from the root to $\square$ do not change the edges between leaves of B (or H). Since there are finitely many monoid elements, there are finitely many possible values for this path.

Let us now consider edges between leaves of B and leaves of $C[\square]$. Similarly as for the leaves in boughs, one can associate to every leaf of $C[\square]$ a *context type* allowing to determine the presence of an edge. For a leaf x in $C[\square]$ that belongs to the root subtree, we collect the following monoid elements: $[\![\mathsf{root} : z]\!]_{\mathcal{T}_{\mathsf{root}}}$, $[\![z : x]\!]_{\mathcal{T}_{\mathsf{root}}}$, and $[\![z : \square]\!]_{\mathcal{T}_{\mathsf{root}}}$, where z is the closest ancestor of x belonging to the path from root to $\square$. For a leaf that belongs to the left (resp. right) subtree, we only consider the monoid element $m_{\mathsf{left}} \cdot [\![\mathsf{root} : x]\!]_{\mathcal{T}_{\mathsf{left}}}$ (resp. $m_{\mathsf{right}} \cdot [\![\mathsf{root} : x]\!]_{\mathcal{T}_{\mathsf{right}}}$). It is immediate that to compute the presence of an edge between a leaf y in B and a leaf x in $C[\square]$, it suffices to know the bough type of y and the context type of x (see **??**, and **??**). Since there are finitely many possible context types and bough types, if a bough B is a good bough in $C[B]$ using a map h and a compatible bough H, then for any other context $C'[\square]$ sharing the same context types for its leaves as $C[\square]$, the bough B is also a good bough in $C'[B]$ using the same map h and the same compatible bough H.

Since there are finitely many possible contexts up to this equivalence relation, we conclude the proof. ◀

Proof of ?? implies ?? of ??. Assume towards a contradiction that the class of graphs obtained as images of $\mathcal{I}$ is 2-well-quasi-ordered, but that there exist bad boughs of arbitrarily large dimension. Let $(B_i)_{i \geq 1}$ be an infinite sequence of such bad boughs. Without loss of generality, we can assume that all boughs B_i have the same level k and are pairwise compatible, by **??**.

Using **??**, we can extract one context $C[\square]$ such that infinitely many boughs B_i are bad boughs in $C[B_i]$. Finally, we can extract further to assume that the dimension of the boughs B_{i+1} is greater than thrice the number of leaves in B_i.

Let us now colour every graph $\mathcal{I}(C[B_i])$ with two labels: $\bullet$ to leaves belonging to B_i, and colour $\circ$ to leaves belonging to $C[\square]$. Because we assume that the image of $\mathcal{I}$ is 2-well-quasi-ordered, we can extract an infinite increasing subsequence, and assume that for all $i < j$, there exists an embedding $f_{i,j} : \mathcal{I}(C[B_i]) \to \mathcal{I}(C[B_j])$ that preserves labels. Let us remark that for all $i < j$, the map $f_{i,j}$ acts as a permutation when restricted to nodes labelled $\circ$, that is of size the number of leaves in $C[\square]$. By a Ramsey argument, we can therefore assume that this permutation is the identity for all $i < j$.

Finally, consider any two indices $i < j$. Because the dimension of B_j is greater than thrice the number of leaves in B_i, there must exist three consecutive bough blocks of B_j that are left untouched by $f_{i,j}$. Furthermore, we know that $f_{i,j}$ is the identity on leaves belonging to $C[\square]$. Hence, the map $f_{i,j}$ witnesses that B_i is a good bough in $C[B_i]$, which is a contradiction. ◀

5 Bounded Clique Width

In this section, we will aim at leveraging the previous results to tackle classes of bounded clique-width. Our first goal is to prove our **??** regarding the images of finite trees under MSO-interpretations. There are two difficulties here: first, we have only dealt with simple MSO-interpretations so far, and second, we have not yet given any decision procedure to check whether the image of a given simple MSO-interpretation is 2-well-quasi-ordered. We first address the second difficulty.

▶ **Definition 25** (Perfect Bough). *Let $\mathcal{T} = C[B]$ be a tree with a bough B of level k. We say that B is a perfect bough if there exists a compatible bough H of level k and a map $h : \mathcal{I}(C[B]) \to \mathcal{I}(C[H])$*

1. *h is an embedding of graphs,*
2. *h is the identity map on leaves of $C[\square]$,*
3. *the bough type of every leaf x in B is the same as the bough type of $h(x)$ in H,*
4. *there exist three consecutive blocks in H whose leaves are left untouched by h.*

An imperfect bough is a bough that is not a perfect bough.

Note that every perfect bough is a good bough, thus every bad bough is an imperfect bough. This allows us to lift **??** to perfect boughs.

▶ **Lemma 26.** *The following are equivalent:*

1. *the image of $\mathcal{I}$ is 2-well-quasi-ordered,*
2. *the image of $\mathcal{I}$ is $\forall$-well-quasi-ordered,*
3. *there exists a bound on the dimension of imperfect boughs.*

Proof. First, if we have a bound on the dimension of imperfect boughs, then we also have a bound on the dimension of bad boughs. Hence, by **??**, the image of $\mathcal{I}$ is $\forall$-well-quasi-ordered.

Conversely, assume that there exist imperfect boughs of arbitrarily large dimension. Then, by an argument similar to the one of **??**, we can construct an infinite labelled antichain in the image of $\mathcal{I}$. The only change is that one also adds the bough type of the leaves as additional labels. This actually follows the original proof technique of [**?**] for bounded linear clique-width. ◀

The important advantage of **??** over **??** is that imperfect boughs are easily recognizable using Monadic Second-Order logic on trees. Indeed, every bough B is actually a (binary) tree with some distinguished nodes that form its backbone. Hence, given a tree $\mathcal{T}$ together with a split $\mathfrak{s}$ and labels on its edges coming from a finite monoid M, one can ask whether this tree with distinguished nodes represents an imperfect bough.

In order to do so, we first need the following **??**, which explains that when replacing a bough B by a compatible bough H, we do not change the part of the graph represented by the leaves of the bough context $C[\square]$. This is because, to determine the presence of edges between leaves of $C[\square]$, one needs only to know the value of the first idempotent element labelling the bough, which does not change when replacing B by a compatible bough H. Consequently, we need focus only on the leaves of B inside of H to construct an embedding from $\mathcal{I}(C[B])$ to $\mathcal{I}(C[H])$ satisfying the conditions of **??**.

▶ **Fact 27.** *Let $\mathcal{T} = C[B]$ be a tree with a bough B of level k, and let H be a bough that is compatible with B. Let $\mathcal{T}' = C[H]$ be the tree obtained by replacing B by H in $\mathcal{T}$. Then, the subgraph of $\mathcal{I}(\mathcal{T})$ induced by the leaves in the context $C[\square]$ equals the subgraph of $\mathcal{I}(\mathcal{T}')$ induced by the leaves in the context $C[\square]$.*

▶ **Lemma 28.** *There exists an* MSO *formula that recognizes imperfect boughs.*

▶ **Corollary 29.** *We can decide whether there are imperfect boughs of arbitrarily large dimension.*

Proof. We use the theory of Cost-MSO formulas [**?**, Section 6]. Let Φ denote the formula of **??** that recognises imperfect boughs. We now consider the following Cost MSO formula:

$$\Psi(N) \triangleq \Phi \land |\{ \text{ dimension of the bough } \}| \geq N$$

This formula takes an input parameter N (a natural number) and decides whether the provided tree with split and labels is an imperfect bough of dimension at most N. Because we can compare the boundedness of Cost-MSO definable functions [**?**, Corollary 16], we can decide whether such a formula can hold for arbitrarily large values of N, effectively answering our question. ◀

We are now ready to prove our main result.

▶ **Theorem ??.** *Let $\mathcal{C}$ be the image of a class of finite trees under an* MSO *interpretation $\mathcal{I}$. Then, the following are equivalent and decidable given $\mathcal{I}$:*

1. *$\mathcal{C}$ is 2-well-quasi-ordered,*
2. *$\mathcal{C}$ is $\forall$-well-quasi-ordered,*
3. *$\mathcal{C}$ is orderably $\forall$-well-quasi-ordered,*

Proof of ??. Let $\mathcal{I}$ be an MSO-interpretation. We can eliminate the domain formula and selection formula by standard arguments without changing the 2-well-quasi-ordering status of the image of $\mathcal{I}$ [?, Lemma 8]. Thus, we can assume that $\mathcal{I}$ is a simple MSO-interpretation. Furthermore, the transformation from a simple MSO-interpretation to a monoid interpretation from trees described in ?? is effective. Finally, by ?? the image of $\mathcal{I}$ is 2-well-quasi-ordered if and only if it is $\forall$-well-quasi-ordered, and by ??, we can decide whether the image of $\mathcal{I}$ is 2-well-quasi-ordered.

It remains to discuss the fact that we can add a total ordering on the graphs without changing the 2-well-quasi-ordering status of the image of $\mathcal{I}$. This is because we prove that the image of $\mathcal{I}$ is 2-well-quasi-ordered using the marked gap-embedding ordering on marked nested trees representing the graphs, which already includes a total ordering on the leaves. ◀

It is straightforward to derive that the same holds for hereditary classes of bounded clique-width, thus proving ??.

▶ **Theorem ??.** *The conclusion of ?? holds when $\mathcal{C}$ is a hereditary class of graphs having bounded clique-width.*

Proof of ??. By a standard argument, if the class $\mathcal{C}$ is hereditary and 2-well-quasi-ordered, then it is characterised by finitely many forbidden induced subgraphs [?, Proposition 3]. Thus, if $\mathcal{C}$ has bounded clique-width and is hereditary, we can assume that $\mathcal{C}$ is exactly the image of an MSO-interpretation $\mathcal{I}$ from finite trees: this is done by ensuring in the formula ϕ_{dom} that the graph that would be produced does not contain any of these forbidden induced subgraphs. The result then follows from ??. ◀

Let us now turn to ??, and in particular the link between 2-well-quasi-ordering and bounded linear clique-width. The following lemma shows that the existence of imperfect boughs of arbitrarily large dimension can be witnessed using only blocks from a finite set.

▶ **Lemma 30.** *Assume that there are imperfect boughs of arbitrarily large dimension. Then there exists a finite set of blocks $\mathbb{F}$ such that we can build an imperfect bough of arbitrarily large dimension using only blocks from $\mathbb{F}$.*

As an immediate consequence of ?? and ??, we can transform long imperfect boughs into long imperfect boughs using only blocks from a finite set. This allows us to prove the following corollary.

▶ **Corollary 31.** *The image of $\mathcal{I}$ is 2-well-quasi-ordered if and only if every subclass of bounded linear clique-width in $\mathcal{I}$ is 2-well-quasi-ordered.*

Proof. If the image of $\mathcal{I}$ is 2-well-quasi-ordered, then every subclass of bounded linear clique-width in $\mathcal{I}$ is also 2-well-quasi-ordered. Conversely, if the image of $\mathcal{I}$ is not 2-well-quasi-ordered, then by ?? there exist imperfect boughs of arbitrarily large dimension. We consider the finite set of blocks $\mathbb{F}$ given by ??, to obtain an infinite sequence of imperfect boughs $(B_i)_{i\in\mathbb{N}}$ of increasing dimension using only blocks from $\mathbb{F}$.

Note that if B is an imperfect bough in some tree $\mathcal{T} = C[B]$, then it remains an imperfect bough in any context $C'[\square]$, as the definition of perfect bough is local to the bough itself. Thus, we can select a single context $C[\square]$ such that B_i is an imperfect bough inside of $C[\square]$ for all $i \in \mathbb{N}$. Similarly as in ??, we can then build an infinite antichain of two-labelled graphs inside of $\mathcal{I}$ by colouring vertices of $\mathcal{I}(C[B_i])$ with $\circ$ if they come from $C[\square]$ and with $\bullet$ if they come from B_i.

Finally, note that the collection of graphs $\mathcal{I}(C[B_i])$ for $i \in \mathbb{N}$ has bounded linear clique-width. Indeed, the context $C[\square]$ is fixed, and the blocks used to build the boughs B_i come

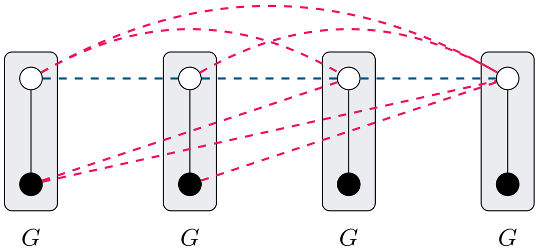

Figure 6 The graph S_4 as described in **??**. The dashed edges represents the edges defined by C and F, respectively in blue and in red. Vertices originating from the same copy of G are grouped in rectangles.

from a finite set $\mathbb{F}$. Consequently, we can represent these graphs by using one colour per node of the context $C[\square]$ plus one colour per node in each block in $\mathbb{F}$, and write single branch (linear) expressions representing all graphs $\mathcal{I}(C[B_i])$. ◀

6 Structural Characterisations

In this section, we will be interested in extracting a very regular structure from a graph class $\mathcal{C}$ that is not $\forall$-well-quasi-ordered. This structure will take three forms, the first one is a combinatorial notion of regular sequence, in continuation with the work of [?] and our **??**. The second one is a notion of periodic sequence that originates from [?, Section 7], and is at the heart of their [?, Conjecture 2]. Those structural notions encode "path-like" behaviour, and we actually manage to extract from those all finite paths using an existential transduction, answering positively to **??**.

▶ **Definition 32.** *A regular sequence (of graphs) is given by a finite graph G, a labelling function $\ell \colon V(G) \to \Sigma$ for some finite set Σ, and two sets $C, F \subseteq \Sigma^2$. It defines an infinite set of graphs $(G^r)_{r \geq 1}$ as follows: the vertex set of G^r is $V(G) \times \{1, \dots, r\}$, and there is an edge between (u, i) and (v, j) with $i \leq j$ if and only if one of the following holds:*

- $(u, v) \in E(G)$ *and* $i = j$;
- $(\ell(u), \ell(v)) \in C$ *and* $|i - j| = 1$
- $(\ell(u), \ell(v)) \in F$ *and* $|i - j| > 1$.

▶ **Example 33.** Let us consider the graph G that is a path with two vertices, respectively labelled $\bullet$ and $\circ$. Let us define $C \triangleq \{(\circ, \circ)\}$ and $F \triangleq \{(\circ, \circ), (\bullet, \circ)\}$, the graph G^n is the *split-permutation graph of size n* that we write S_n and can be seen in **??**.

A *regular antichain* is a regular sequence that is an antichain for the labelled induced subgraph relation when vertices of G^r are labelled by their color in G, and the first and last copies of G in G^r are additionally labelled by two distinct new labels $\triangleright$ and $\triangleleft$. It follows results from [?] and our previous **??** that such structures can be found in non-$\forall$-well-quasi-ordered hereditary classes of graphs of bounded clique-width.

▶ **Lemma 34.** *Let $\mathcal{C}$ be a hereditary class of graphs of bounded clique-width that is not $\forall$-well-quasi-ordered. Then, it contains a regular antichain.*

Our next goal is to show that can existentially transduce all finite paths from a regular antichain. Remark that split-permutation graphs form a regular antichain, but exclude P_4

as an induced subgraph. Before that, let us briefly recall that existentially transducing all paths is an obstruction to being $\forall$-well-quasi-ordered.

▶ **Lemma 35.** *Let $\mathcal{C}$ be a hereditary class that existentially transduces all paths, then $\mathcal{C}$ is not $\forall$-well-quasi-ordered.*

Proof. Let $\mathcal{I} = (\phi_{\text{univ}}, \phi_{\text{edge}}, \phi_{\text{dom}})$ be an existential interpretation from $\mathsf{Label}_\Sigma(\mathcal{C})$ to the class of finite graphs, such that the image of $\mathsf{Label}_\Sigma(\mathcal{C})$ contains all finite paths. Let us write G_i for a graph in $\mathsf{Label}_\Sigma(\mathcal{C})$ such that $\mathcal{I}(G_i) = P_i$, where P_i is the path on i vertices. Remark that if $G_i \subseteq_i G_j$ as labelled graphs, then $P_i = \mathcal{I}(G_i)$ is a subgraph (not necessarily induced) of $\mathcal{I}(G_j) = P_j$. Indeed, whenever $\phi_{\text{univ}}(u)$ holds in G_i, it also holds in G_j, because G_i is an induced subgraph of G_j and ϕ_{univ} is existential. Similarly, whenever $\phi_{\text{edge}}(u, v)$ holds in G_i, it also holds in G_j.

Let us consider an extra labelling information on the vertices of the graphs in $\mathsf{Label}_\Sigma(\mathcal{C})$, where we color the two endpoints of each path P_n with two distinct colors $\triangleright$ and $\triangleleft$. Using these new labels, one can existentially interpret the class of all finite paths with endpoints colored with $\triangleright$ and $\triangleleft$.

Assume towards a contradiction that $\mathcal{C}$ is $\forall$-well-quasi-ordered. Then, there is a pair $i < j$ such that $G_i \subseteq_i G_j$ as labelled graphs. By the previous remark, this implies that P_i (with coloured endpoints) is a labelled subgraph of P_j (with coloured endpoints), which is impossible. ◀

▶ **Lemma 36.** *Let $\mathcal{C}$ be a regular antichain. Then $\mathcal{C}$ existentially transduces the class of all finite paths.*

Proof sketch. Let $\mathcal{C} = (G^r)_{r\geq 1}$ and let $\ell \colon V(G) \to \Sigma$ be its labelling function with sets $C, F \subseteq \Sigma^2$. Without loss of generality, assume that ℓ is injective up to duplicating labels from Σ, and assume that $|\Sigma| = |V(G)|$. We define the following quantifier-free formula that uses the edge predicate on G^r and the labelling function $\ell \colon V(G^r) \to \Sigma$.

$$\phi_{\to}(u, v) = \neg(E(u, v) \Leftrightarrow (\ell(u), \ell(v)) \in F) \quad .$$

Informally, $\phi_{\to}(u, v)$ creates an arc (u, v) if the edge relation between $u, v \in G^r$ differs if u and v are put far away form each other in this order.[6]

To obtain a path, we first show that one can obtain a long directed shortest path from the first copy of G to the last copy of G, going through every copy of G. Then, from this so-called *spanning path*, we extract undirected paths by analysing the behaviour of potential "backward edges" along the directed path using existential formulas built upon $\phi_{\to}$. ◀

Finally we provide a last structural characterisation defined in [?, Section 7] that is not $\forall$-well-quasi-ordered similar to periodic sequence of graphs.

▶ **Definition 37** ([?, Section 7]). *A periodic sequence (of graphs) is given by a finite word w over a finite alphabet Σ, and two sets $C, F \subseteq \Sigma^2$. It defines an infinite set of graphs $(G_{w^r})_{r\geq 1}$ as follows: the vertex set of G_{w^r} is $\{1, \dots, r \cdot |w|\}$, and there is an edge between $i \leq j$ if*

- *$i + 1 = j$ and $((w^r)_i, (w^r)_j) \in C$,*
- *$i + 1 < j$ and $((w^r)_i, (w^r)_j) \in F$.*

▶ **Example 38.** Let us consider the word $w = \circ\bullet$, and $C \triangleq \{(\circ, \bullet)\}$ and $F \triangleq \{(\circ, \circ), (\bullet, \circ)\}$. The graph G_{w^n} is the split-permutation graph of size n.

[6] It could be that u can be in a copy of G that comes after the one of v in G^r.

It is immediate to see that periodic sequences are a special case of regular sequences, by grouping vertices of w into a single graph G. A *periodic antichain* is a periodic sequence that is an antichain for the labelled induced subgraph relation when vertices of G_{w^r} are not labelled, except for the first and last vertices respectively labelled by $\triangleright$ and $\triangleleft$. In the original definition of [?], it was required by periodic sequences to have $C = \Sigma^2 \setminus F$. It is easy to verify that with this condition a periodic sequence is an antichain if the first and last vertices are labelled.

While it is not immediate from the definitions that one can turn a regular antichain into a periodic antichain, it follows from our proof of **??** that when we existentially transduce all paths from a regular antichain, we actually isolate a periodic antichain.

▶ **Corollary 39.** *Let $\mathcal{C}$ be a hereditary class, then $\mathcal{C}$ contains a regular antichain if and only if it contains a periodic antichain.*

▶ **Theorem ??.** *Let $\mathcal{C}$ be a hereditary class of finite graphs of bounded clique-width. Then, the following are equivalent:*

1. *$\mathcal{C}$ is not 2-well-quasi-ordered,*
2. *There exists a class $\mathcal{D} \subseteq \mathcal{C}$ of bounded linear clique-width which is not 2-well-quasi-ordered,*
3. *There exists a regular antichain in $\mathcal{C}$,*
4. *There exists a periodic antichain in $\mathcal{C}$,*
5. *$\mathcal{C}$ existentially transduces the class of all finite paths.*

Proof of ??. First, **??** implies **??** by **??** and the fact that 2-well-quasi-orderings and $\forall$-well-quasi-orderings coincide in our setting thanks to **??**.

Then, **??** $\iff$ **??** was obtained by our analysis in **??**. This analysis was refined to prove that **??** implies **??** in **??**. Then, we proved in **??** that one can existentially transduce paths from a regular antichain, and therefore, we obtain **??** implies **??**.

Finally, the equivalence of **??** and **??** was obtained in **??**. ◀

7 Concluding Remarks

We have proven that for hereditary classes of graphs of bounded clique-width, being $\forall$-well-quasi-ordered is a very robust notion, that enjoys many structural characterisations, and where most conjectures from the literature on labelled well-quasi-orderings hold true. We consider these results as a first step towards understanding the landscape of labelled well-quasi-orderings of graphs and other relational structures.

A natural research direction would be to consider other classes of relational structures. We strongly believe that our techniques can be adapted to the case of finite relational structures over a finite relational signature. Another interesting direction would be to consider non-hereditary classes of graphs of bounded clique-width. In this case, we also believe that our techniques can be adapted.

A fundamental limitation of our work is that we assume classes to be of bounded clique-width. We believe that classes of graphs that are 2-well-quasi-ordered are necessarily of bounded clique-width (see **??**), and a first step would be to show that every such class is *monadically dependent*.

Another question left unanswered by our work is whether we can existentially interpret all paths in every hereditary class of graphs that is not $\forall$-well-quasi-ordered (and of bounded clique-width). An argument similar to **??** shows that if a class existentially interprets all paths, then it is not 2-well-quasi-ordered, but the converse direction remains open.

Finally, note that we proved a very strong dichotomy result for hereditary classes of graphs of bounded clique-width: either they are not 2-well-quasi-ordered or the induced subgraph relation on them is simpler than the gap-embedding relation on trees. We conjecture that this dichotomy extends to all hereditary classes of graphs: there is nothing beyond (nested) trees.

path we obtain an infinite antichain. Moreover, being ℓ-wqo entails being k-wqo for all $k \leq \ell$ and similarly being n^--wqo entails being m^--wqo for all $m \leq n$. Finally, if $\mathcal{C}$ is 2-wqo then it is n^--wqo for all n [?, Lemma 5.2].

▶ **Conjecture 40.** *Every hereditary class of graphs that is well-quasi-ordered by the induced subgraph relation is monadically dependent.*

▶ **Conjecture 41.** *Every hereditary class of graphs that is 2^--well-quasi-ordered by the induced subgraph relation is 2-well-quasi-ordered by the induced subgraph relation.*

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A Proofs of ??

▶ **Fact ??.** *Let $\mathcal{T}_1, \mathcal{T}_2$ be two trees and $\mathfrak{s}_1, \mathfrak{s}_2$ be two forward Ramseyan splits of height N of $\mathcal{T}_1, \mathcal{T}_2$ Assume that $h\colon \mathcal{T}_1 \to \mathcal{T}_2$ is a gap-embedding from $(\mathcal{T}_1, \mathfrak{s}_1)$ to $(\mathcal{T}_2, \mathfrak{s}_2)$. Then, for every u, v in $\mathcal{T}_1$, if $\mathfrak{s}_1(u : v) > k$ and $\mathfrak{s}_1(v) = k$ for some $k \in \{1, \dots, N\}$, then $\mathfrak{s}_2(h(u) : h(v)) \geq k$.*

Proof of ??. We prove this by induction on the length of the path from u to v in $\mathcal{T}_1$. The base case is exactly **??** of the definition of gap-embedding.

For the inductive case, let u, v be two nodes in $\mathcal{T}_1$ such that $\mathfrak{s}_1(u : v) > k$ and $\mathfrak{s}_1(v) = k$ for some $k \in \{1, \dots, N\}$. Let w be the parent of v in $\mathcal{T}_1$. Since $\mathfrak{s}_1(u : v) > k$, we have $\mathfrak{s}_1(w) > k$. By the induction hypothesis, we have $\mathfrak{s}_2(h(u) : h(w)) \geq k$. Furthermore, by definition of gap-embedding, we have $\mathfrak{s}_2(h(w) : h(v)) \geq k$. Hence $\mathfrak{s}_2(h(u) : h(v)) \geq k$. ▷ Back to p.?? ◀

▶ **Theorem ??.** *For every finite L, the class of L-bounded well-marked nested trees labelled over a well-quasi-ordered set $(X, \leq_X)$ is well-quasi-ordered under the gap-embedding ordering.*

Proof of ??. We define a function f that maps L-bounded well-marked nested trees to nested trees labelled over a well-quasi-ordered set, such that if there exists a gap-embedding between $f(\mathcal{T}_1, \mathfrak{s}_1, \rho_1)$ and $f(\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$, then $\mathcal{T}_1 \leq_{\text{gap}} \mathcal{T}_2$ as marked nested trees. Using **??**, this concludes the proof.

We can encode **??????????** of **??** by adding suitable labels to the nodes of the trees. Formally, we define a new labelling of the nodes of a tree $\mathcal{T}$ as follows: we add to each node x of $\mathcal{T}$ the label $[\![z : x]\!]_{\mathcal{T}}$ for every $k \in \{1, \dots, N\}$, where z is the least ancestor of x such that $\mathfrak{s}(z) = k$, or a special symbol $\perp$ if no such ancestor exists. Furthermore, if y is the immediate left (respectively right) child of x in $\mathcal{T}$, we also add the label $[\![x : y]\!]_{\mathcal{T}}$ to x, or the special symbol $\perp$ if no such child exists. We distinguish the root by adding a special label root, and we add a special label leaf to every leaf. Finally, we also add the label $\rho(x)$ to x. Since M is finite, the new labelling is still over a well-quasi-ordered set.

It remains to ensure that the mapping h is gluing on non-dummy nodes. To that end, we modify the split $\mathfrak{s}$ into a new split $\mathfrak{s}'$ as follows: for every non-dummy node x, we have that x is the ith element in a maximal path of non-dummy nodes $x_1 \sqsubset_{\mathcal{T}} x_2 \sqsubset_{\mathcal{T}} \cdots \sqsubset_{\mathcal{T}} x_n$ that contains x. Because the tree is L-bounded, we have $n \leq L$. We set $\mathfrak{s}'(x) = N + 1 + i$. For a dummy node y, we set $\mathfrak{s}'(y) = \mathfrak{s}(y)$.

Note that any gap-embedding h from $(\mathcal{T}_1, \mathfrak{s}'_1)$ to $(\mathcal{T}_2, \mathfrak{s}'_2)$ is necessarily gluing on non-dummy nodes. Assume that $\rho_1(y) \in \{\mathbf{M}, \mathbf{S}\}$, and let x be the parent of y in $\mathcal{T}_1$. Since $\mathfrak{s}'_1(y) = N + 1 + i$, we know that $\mathfrak{s}'_2(h(x)\colon h(y)) \geq \mathfrak{s}'_1(y) = N + 1 + i$. This means that between $h(x)$ and $h(y)$ there are only nodes with split depth greater or equal than $N + 1 + i$. If $i = 1$ (i.e., y is the first node of a maximal path of non-dummy nodes), then there is no such node, and $h(x)$ is the parent of $h(y)$. Otherwise, $\mathfrak{s}'_1(x) = N + i > N$, and therefore the path from $h(x)$ to $h(y)$ contains only non-dummy nodes. Since the split values have been chosen to be strictly increasing inside maximal paths of non-dummy nodes, there is no such node, and $h(x)$ is the parent of $h(y)$. ▷ Back to p.?? ◀

B Proofs of ??

▶ **Lemma ??.** *Let k be a level. There are finitely many boughs of level k up to compatibility.*

Proof of ??. Let $\mathcal{T} = C[B]$ be a tree obtained by plugging a bough B of level k inside a context $C[\square]$. Our aim is to extract finitely many parameters from B (independently of

$C[\square]$) such that any other bough B' of level k with the same parameters is compatible with B. These parameters will be elements of M or tuples thereof, and since M is finite, this immediately gives finiteness.

To that end, fix a context $C[\square]$. Assume that $\mathcal{T} = C[B]$ is forward Ramseyan. We want to ensure that $\mathcal{T}' = C[B']$ is forward Ramseyan for any B' with the same parameters. Recall that $\mathcal{T}'$ is forward Ramseyan if, for every branch in $\mathcal{T}'$, every level l, every l-neighbourhood in this branch, and every four nodes $x \sqsubset y$ and $x' \sqsubset y'$ in this l-neighbourhood, we have $[\![x : y]\!]_{\mathcal{T}'} \cdot [\![x' : y']\!]_{\mathcal{T}'} = [\![x : y]\!]_{\mathcal{T}'}$.

First, remark that l-neighbourhoods with $l < k$ inside $\mathcal{T}'$ are fully contained either in B', or $\mathcal{T}_{\mathsf{root}}$, or $\mathcal{T}_{\mathsf{left}}$, or $\mathcal{T}_{\mathsf{right}}$. As a consequence, nothing needs to be checked on such l-neighbourhoods since B' is assumed to be locally forward Ramseyan. More generally, the only possible failures of the forward Ramseyan condition in $\mathcal{T}'$ must involve nodes that are not all contained in one of these four subtrees.

Now, consider l-neighbourhoods with $l \geq k$. In that case, the only l-neighbourhood of a branch that is not fully contained in one of the four subtrees crosses B'. Let us distinguish several cases based on the shape of the branch with respect to the context $C[\square]$ and the bough B'.

1. The branch containing the l-neighbourhood starts in $\mathcal{T}_{\mathsf{root}}$ (possibly in $\square$), goes through B', and ends in $\mathcal{T}_{\mathsf{left}}$ or $\mathcal{T}_{\mathsf{right}}$.
2. The branch containing the l-neighbourhood starts in $\mathcal{T}_{\mathsf{root}}$ (possibly in $\square$), goes through B', and ends in B'.

In the first case, consider two nodes $x \sqsubset y$ and $x' \sqsubset y'$ in the l-neighbourhood. The value $[\![x : y]\!]_{\mathcal{T}'}$ can be decomposed into a part before entering B', a part inside B', and a part after leaving B'. By construction, the part inside B' is one of the idempotent elements labelling the backbone of B'. Hence, to ensure that $\mathcal{T}'$ is forward Ramseyan in this case, it suffices to check finitely many equations involving those idempotent elements (note that the number of equations depends on the shape of $C[\square]$, but not on the number of elements of M extracted from B').

In the second case, consider two nodes $x \sqsubset y$ and $x' \sqsubset y'$ in the l-neighbourhood. Here, the value $[\![x : y]\!]_{\mathcal{T}'}$ can be decomposed into a part before entering B', a part from b'_0 to the first node z in B' of level l (which is not on the backbone of B' if $l > k$), and a part from z to y. Similarly, the equations ensuring that $\mathcal{T}'$ is forward Ramseyan in this case can be expressed using the values $[\![b'_0 : z]\!]_{\mathcal{T}'}$ for every node z in B', and the values $[\![z : y]\!]_{\mathcal{T}'}$ for every node y in B' in the same l-neighbourhood as z (for some branch). Note that one only cares about the existence of pairs of values $([\![b'_0 : z]\!]_{\mathcal{T}'}, lbl\mathcal{T}'zy)$ for nodes z and y in the same l-neighbourhood, and there are only finitely many such pairs.

We have proved that, assuming that $\mathcal{T} = C[B]$ is forward Ramseyan, it suffices to check finitely many parameters of B' to ensure that $C[B]$ is forward Ramseyan. Since M is finite, there are finitely many possible values for these parameters, which concludes the proof. ▷

`Back to p.??` ◀

C Proofs of ??

▶ **Lemma ??.** *There exists an* MSO *formula that recognizes imperfect boughs.*

Proof of ??. The proof follows the same pattern as [**?**, Lemma 19], but adapts it from words to trees. Assume that B is a perfect bough of level k inside a tree $\mathcal{T} = C[B]$. This gives us a compatible bough H of level k and an embedding $h\colon \mathcal{I}(C[B]) \to \mathcal{I}(C[H])$ satisfying the conditions of **??**.

Consider five copies of B, called B_{left}, $B_{\mathsf{mid}-1}$, $B_{\mathsf{mid}-2}$, $B_{\mathsf{mid}-3}$, and B_{right}. Define B^5 to be the bough obtained by concatenating these five copies, merging b_n of B_{left} with b_0 of $B_{\mathsf{mid}-1}$, merging b_n of $B_{\mathsf{mid}-1}$ with b_0 of $B_{\mathsf{mid}-2}$, and so on. We claim that B and B^5 are compatible boughs.

Define a mapping $k\colon \mathcal{I}(C[B]) \to \mathcal{I}(C[B^5])$ as follows: leaves x of B are mapped to their corresponding leaf in B_{left} or B_{right} depending on whether $h(x)$ is to the left or right of the untouched bough blocks of H. For leaves x of $C[\square]$, we let $k(x) = x$. Thanks to **??**, we know that k preserves edges between leaves of $C[\square]$. Furthermore, because h is an embedding that preserves the bough type of leaves in B, and because bough types suffice to compute the presence of an edge between a leaf of B and a leaf of $C[\square]$ (or between two leaves of B), we deduce that k is also an embedding.

We have therefore shown that if B is a perfect bough, then there exists a mapping k' from leaves of B to $\{\mathsf{left}, \mathsf{right}\}$ such that there is an edge between two leaves x and y of B in the graph $\mathcal{I}(C[B])$ if and only if the bough types of x and y are such that there is an edge between $x_{k'(x)}$ and $y_{k'(y)}$ in the graph $\mathcal{I}(C[B^5])$. One can guess such a mapping k' in Monadic Second-Order logic, and then verify the required conditions on edges between leaves of B. As a consequence, one can write an MSO formula that holds if and only if B is a perfect bough. ▷ Back to p.?? ◀

▶ **Lemma ??.** *Assume that there are imperfect boughs of arbitrarily large dimension. Then there exists a finite set of blocks $\mathbb{F}$ such that we can build an imperfect bough of arbitrarily large dimension using only blocks from $\mathbb{F}$.*

Proof of ??. The proof follows the same pattern as **????**, and continues along the ideas of **??**. The idea is that if a bough B is perfect, then one can witness it by using the compatible bough B^5 defined by gluing five copies of B together, because we observed that perfectness essentially depends on the bough types of the leaves of B. Another consequence of this analysis is that if one defines an equivalence relation between bough blocks of a given bough B based on the set of available bough types of its leaves, then one can always replace a bough block by another one in the same equivalence class without changing the perfectness of B. We conclude because this equivalence relation has finitely many classes. ▷ Back to p.?? ◀

D Proofs of ??

▶ **Lemma ??.** *Let $\mathcal{C}$ be a hereditary class of graphs of bounded clique-width that is not $\forall$-well-quasi-ordered. Then, it contains a regular antichain.*

Proof of ??. Using **??**, one may assume that $\mathcal{C}$ has bounded linear clique-width. It then follows from the analysis of [?] that $\mathcal{C}$ contains a periodic sequence $(G^r)_{r\geq 1}$. We restate the argument here for completeness.

Assume that $\mathcal{C}$ is not $\forall$-well-quasi-ordered. We proceed as in the proof of **??**, and assume without loss of generality that $\mathcal{C}$ is the image of a monoid interpretation. Using **??** and **??**, we obtain one context $C[\square]$, a finite set $\mathbb{F}$ of blocks, and a sequence $(B_i)_{i\in\mathbb{N}}$ of boughs built using blocks from $\mathbb{F}$ and such that for every $i \in \mathbb{N}$, B_i is an imperfect bough in the context $C[\square]$.

We can encode B_i as a word w_i on the finite alphabet $\mathbb{F}$. As in **??**, we can construct an MSO formula that decides whether a word $w \in \mathbb{F}^*$ encodes an imperfect bough in the (fixed) context $C[\square]$. Because the language $L_{\text{imperfect}}$ of such words is regular and infinite, there exist three words u, v, w in $\mathbb{F}^*$ such that $uw^nv \in L_{\text{imperfect}}$ for all $n \in \mathbb{N}$.

From w, one can create the desired graph G, and the sets F and C. The vertices of G are the leaves of the bough blocks composing w, and edges are exactly those prescribed by the interpretation $\mathcal{I}$ on $C[w]$. An edge between vertices in two distinct copies of w among vertices of w^n in the graph $\mathcal{I}(C[uww^nwv])$ depends on the order of the vertices, and on whether the two copies are adjacent because of the forward Ramseyan property. We use as colours for the vertices of G their bough types and their precise location in the word w. Using this finite information, one can construct F and C such that G^n is the induced subgraph of $C[uww^nwv]$ obtained by considering vertices from the inner w^n part.

We have extracted a regular sequence of graphs that belongs to $\mathcal{C}$ because the latter is a hereditary class, and because $\mathcal{C}$ is the image of $\mathcal{I}$.[7]

Let us now prove that it is in fact a regular antichain. Assume for contradiction that it is not. Then, without loss of generality, there exists a labelled embedding h from G^n into G^ℓ with $n < \ell$, such that

1. h acts as the identity from the first copy of G in G^n to the first copy of G in G^ℓ,
2. h acts as the identity from the last copy of G in G^n to the last copy of G in G^ℓ,
3. h leaves three consecutive copies of G untouched in G^ℓ.

One can extend h to be the identity on the context $C[uw\square wv]$ and conclude that uww^nwv is a perfect bough in the context $C[\square]$. This is absurd. ▷ Back to p.?? ◀

▶ **Lemma ??.** *Let $\mathcal{C}$ be a regular antichain. Then $\mathcal{C}$ existentially transduces the class of all finite paths.*

Proof of ??. Let $\mathcal{C} = (G^r)_{r\geq 1}$ and let $\ell \colon V(G) \to \Sigma$ be its labelling function with sets $C, F \subseteq \Sigma^2$.

Without loss of generality, assume that ℓ is injective up to duplicating labels from Σ, and assume that $|\Sigma| = |V(G)|$. In the rest of this proof, we assume that $r \geq 3$. For $i \leq r$, we denote the ith copy of G in G^r by G_i, and by $u_i \in G^r$ the ith copy of $u \in V(G)$. We proceed in two steps: first we define a formula producing a directed graph where there is a directed shortest path from G_1 to G_r going through every G_i, but there may exist arcs going from G_i to G_j for $j \leq i$. We then show how to handle these backward arcs, and produce arbitrarily long paths.

We define the following quantifier-free formula that uses the edge predicate on G^r and the labelling function $\ell \colon V(G^r) \to \Sigma$.

$$\phi_{\to}(u,v) \triangleq \neg(E(u,v) \Leftrightarrow (\ell(u),\ell(v)) \in F) \quad . \tag{3}$$

Informally, $\phi_{\to}(u,v)$ creates an arc (u,v) if the edge relation between $u,v \in G^r$ differs if u and v are put far away form each other in this order.[8]

Let $H^r = \mathcal{J}(G^r)$ where $\mathcal{J} \triangleq (\top, \phi_{\to}, \top)$ as a quantifier-free interpretation. Note that H is a directed graph as $\phi_{\to}$ is not symmetric. However, we will show how to existentially interpret paths from the resulting directed graph.

First, let us observe that in H^r there is no arc from a copy i to a copy $j > i+1$.

▷ Claim 42. For $(u_i, v_j) \in E(H^r)$ only if $j \leq i+1$.

Proof. If $j \geq i+1$, then $(u,v) \in E(G^r)$ if and only if $(l(u), l(v)) \in F$ by definition of G^r. ◁

[7] Otherwise, the graphs $\mathcal{I}(C[uww^nwv])$ may not belong to $\mathcal{C}$.
[8] It could be that u can be in a copy of G that comes after the one of v in G^r.

Then, let us remark that if there is a backward arc from the i-th copy to the j-th copy with $j < i$, then for every other previous copy, this arc also exist.

▷ Claim 43. If $(u_i, v_j) \in E(H^r)$ for $j < i$ then for every $j' \leq i$, $(u_i, v_{j'}) \in E(H^r)$.

A *spanning path* of H^r is a directed shortest path from $V(H_1)$ to $V(H_r)$ going through at least one vertex from each $V(H_i)$. The following claim, show that a spanning path must exist in H^r because G^r is a regular antichain.

▷ Claim 44. There is a spanning path in H^r.

Proof. Let V_i^+ be the vertices of H_i such that there is an arc from H_{i-1} to them, V_i^- be the vertices of H_i such that there is an arc from them to H_{i+1}, and V_i^o be the remaining vertices of $V(H_i)$.

Suppose there is no spanning path in H^r, by **??**, it has to be because, for some i, either there is no path in H_i from V_i^+ to V_i^-, or that they are empty (i.e. there is no arc from H_i to H_{i+1}, in which case $V_i^o = V(H_i)$).

Let C_i^+ (resp. C_i^-) be the vertices of H_i that can be reached from V_i^+ (resp. that can reach V_i^-), and C_i^o the remaining vertices of H_i.

We now argue that if there is no spanning path, then one can embed G^r into $G^{r+2+\ell}$ for any $\ell \geq 0$. Fix $1 < i < r$, and define the mapping h as follows: map the first $i-1$ copies of G in G^r and V_i^+ and V_i^o to the corresponding first i copies in G^{r+2}; V_i^- is mapped to the $i+2+\ell$-th copy of G in $G^{r+2+\ell}$; and the last $r-i$ copies of G in G^r are mapped to the last $r-i$ copies of G in $G^{r+2+\ell}$.

It is clear from construction that the middle part of $G^{r+2+\ell}$ is left untouched by this mapping, and that the first and last copies of G in G^r are mapped to the first and last copies of G in $G^{r+2+\ell}$.

Assume towards a contradiction that that this is not an embedding of the underlying graphs, then it means that there is $u \in V_i^+ \cup V_i^o$ and $v \in V_i^-$ such that $(u, v) \in E(G^r)$ and $(h(u), h(v)) \notin E(G^{r+2})$ or the reverse. However as $(h(u), h(v))$ is entirely defined by $(\ell(u), \ell(v))$ and F, it means that $\phi_\to(u, v)$ is satisfied and $(u, v) \in E(H^r)$. Hence if there is no spanning path, then $(G^r)_{r \geq 1}$ is not a regular antichain. ◁

Assuming $r > |V(G)|$, there are two vertices u_i and u_j, $i < j$, in the spanning path of H^r, which are copies of the same vertex $u \in V(G)$. Take these two vertices to be such that the distance d from u_i to u_j is the smallest possible in H^r. Let $t = j - i$, this is the period of P, i.e. the number of copies of G it has to go through until it comes back to a copy of the same vertex. Let $Q = u^1, u^2, \ldots, u^d, u^1$ be the vertices corresponding to the period. In particular, Q induces a spanning path of H^{t+1} (assuming u^i is mapped to the correct copy) and Q^r induces a spanning path of $H^{t \times r+1}$. Here Q^r denotes Q repeated r times where the ending u^1 and beginning u^1 of two consecutive repetition are identified.

Let us do a case analysis based on the presence of backward edges on the spanning path represented by Q^r. If there are no such edges, then the directed path can be turned into an undirected path by considering the symmetrisation of $\phi_\to$ and taking induced subgraph (which can be done by adding a vertex-selecting formula and $|\Sigma|$ selecting colours), hence existentially transduce all finite paths from $\mathcal{C}$.

For the difficult case, let us assume that there exists a backward arc $(u, v) \in E(H^r)$ along the spanning path with $u \in P_i$ and $v \in P_j$ with $i > j$, then by **??** this arc exists between every copy of u and v where v is an earlier copy than the one of u. In particular this means that for every u, v such that v appears earlier on the spanning path, $\mathrm{dist}_{H^r}(u, v) \leq 2d$ (recall

d is the length of Q):

$$\mathrm{dist}_{H^r}(u,v) \le \mathrm{dist}_{H^r}(u,x) + 1 + \mathrm{dist}_{H^r}(y,v),$$

where $(x,y) \in E(H^r)$ is a backward arc such that x appear after u and y before v.

We are now ready to transduces paths in the difficult case. Consider H^{2tr+1}, let u be the beginning of Q and $c = \ell(u)$. To guarantee that the distance between two consecutive copies of u in Q^{2tr} is $2d$ we recolour every other copy of u along Q^{2tr} by a dummy colour c'. Furthermore recolour every copy of u not in Q^{2tr} with the same colour. Define now the formula

$$\phi_{\mathrm{Path}}(u,v) \triangleq \wedge \begin{cases} \mathrm{dist}_{H^r}(u,v) \le 2d \\ \mathrm{dist}_{H^r}(v,u) \le 2d \\ \ell(u) = \ell(v) = c \end{cases} . \tag{4}$$

Notice that this formula only uses existential quantifier (recall that $\phi_{\to}$ which is used to define dist_{H^r} is quantifier-free). This formula create a path on the vertices $u^1, u^3, u^5, \ldots, u^{2tr+1} \in Q^{2tr}$, moreover this path is undirected as $\phi_P(u,v)$ is symmetric. The interpretation $\mathcal{K} \triangleq (\top, \phi_{\mathrm{Path}}, \phi_{\mathrm{univ}})$, where $\phi_{\mathrm{univ}}(u) \triangleq l(u) = c$ (recall that only $u^1, u^3, u^5, \ldots, u^{2tr+1}$ are the only vertices with colour c).

In particular $\mathcal{K}(G^{2tr+1}) = P_r$ where P_r is the path on r vertices, and the conclusion follows. ▷ Back to p.?? ◀

▶ **Corollary ??.** *Let $\mathcal{C}$ be a hereditary class, then $\mathcal{C}$ contains a regular antichain if and only if it contains a periodic antichain.*

Proof of ??. We already observed that periodic antichains are regular antichains. For the converse direction, observe that the sequence of vertices obtained from spanning path **??** correspond to a periodic antichain, with Σ being the labels used on vertices of the path, set F defined according to G, and $C = \Sigma^2 \setminus F$. ▷ Back to p.?? ◀